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CHAPTER 3

FUNDAMENTALS AND APPLICATIONS OF QUANTUM-CONFINED STRUCTURES

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Atoms on a small scale behave like nothing on a large scale, for they satisfy the laws of quantum mechanics. So, as we go down and fiddle around with the atoms down there, we are working with different laws, and we can expect to do different things. We can use some system involving the quantized energy levels.

Richard P. Feynman, December 1959.

3.1 Introduction

In this chapter, we consider solar power either in the form of electricity produced by photovoltaic cells or in the form of stored chemical potential in liquid or gaseous fuels. For the latter, examples are hydrogen, alcohol, or hydrocarbons produced using solar photons to split water into its elements or to drive the reaction of carbon dioxide and water to yield reduced carbon compounds and oxygen (*i.e.* artificial photosynthesis). The cost of solar power is determined by the conversion efficiency (η) of the solar converter and its total capital cost per unit area. The ratio of the capital cost per unit area ($\$/\text{m}^2$) to the peak power (W_p) produced per unit area yields the generated solar power cost in units of $\$/\text{peak watt}$ ($\$/W_p$), where W_p is the peak solar power density (watts/unit area) delivered by the device at the surface of the Earth at high noon with no clouds ($W_p = \eta \times 1000 \text{ W/m}^2$). To convert $\$/W_p$ to $\$/\text{kWh}$ an approximate conversion is simply to multiply $\$/W_p$ by a factor F that ranges from 0.03 (www.solarbuzz.com) to 0.05 (Surek, 2005) and takes into account the total insolation per year at a given site on Earth (*i.e.* integrated irradiance over the diurnal cycles and historical cloud cover), interest rates, the system lifetime, depreciation, operating costs, and maintenance; here we use an average value of $F = 0.04 \text{ kWh}/W_p$.

To achieve massive utilisation of solar power on the terawatt (TW) scale (the total energy consumed globally in 2006 was about 13 TW-yr), it is necessary to develop solar conversion systems that are cost-competitive not only with the present cost of electricity derived from coal and nuclear energy, but also with liquid and gaseous fuels derived from fossil sources. In 2007, electricity cost around 0.07–0.10 \$/kWh and the average cost of gaseous and liquid fuel produced from natural gas and petroleum is between 7–14 \$/MMBtu (0.02–0.04 \$/kWh) (see www.eia.doe.gov). The present cost of photovoltaic systems ($\sim$ \\$5.00 for the PV module plus $\sim$ \\$3.00 for the balance-of-systems cost) is $\sim$ 7–9 $\$/W_p$, which is equivalent to about $\$8/W_p \times 0.04 \text{ kWh}/W_p = 0.32 \text{ \$/kWh}$. Thus, to achieve a solar power cost competitive with fossil-derived liquid and gaseous fuels, as well as fossil or nuclear electricity, the cost needs to be in the range of 0.02–0.04 $\$/kWh$ or $\sim 0.03/F \sim \$0.75/W_p$ for the system (and $\sim \$0.40/W_p$ for the module); for PV this represents greater than a tenfold decrease in cost from present-day (2007) systems. To meet this goal, the conversion system has to have both high efficiency and low cost/unit area. Thus, for example, if the total system areal cost can be reduced to $\$150/m^2$ from the present cost of $\sim \$800/m^2$, then the module conversion efficiency needs to be 37.5%. Since the module efficiency is about 75% of the individual cell efficiency, the cell efficiency needs to be about 50% for an areal cost of $\$150/m^2$.

As is well known, the maximum thermodynamic efficiency or so-called ‘detailed balance limit’ for the conversion of unconcentrated solar energy into electrical free energy in a single-threshold absorber was calculated by Shockley and Queisser (1961) to be about 31%; this analysis is also valid for the conversion of unconcentrated solar energy to chemical free energy (Ross, 1966 and 1967). This calculation was based on several constraints: (1) the only loss of photogenerated electrons and positive holes (charge carriers) is through radiative recombination (this is termed the radiative limit); (2) detailed balance is assumed; (3) a maximum yield of one electron–hole pair per absorbed photon is produced; and (4) thermal equilibrium is attained between electrons and phonons. The 31% maximum efficiency value is attainable in semiconductors with bandgaps ranging from about 1.2 eV to 1.4 eV. However, the solar spectrum contains photons with energies ranging from about 0.5 eV to 3.5 eV. Photons with energies below the semiconductor bandgap are not absorbed, while those with energies above the bandgap create electrons and holes with a total excess kinetic energy equal to the difference between the photon energy and the bandgap. This excess kinetic energy creates an effective temperature for a Boltzmann ensemble of the photogenerated carriers that can be much higher than the lattice temperature after the electrons and holes have equilibrated (thermalised) among themselves. Such carriers are termed ‘hot electrons and hot holes’, and the initial carrier temperature on

photon absorption can be as high as 3000 K with the lattice temperature at 300 K. In bulk semiconductors, the division of this kinetic energy between electrons and holes is determined by their effective masses, with the carrier having the lower effective mass receiving more of the excess energy (Nozik, 2001b). Thus

$$\Delta E_e = (h\nu - E_g) \left[1 + m_e^* / m_h^* \right]^{-1} \quad (3.1)$$

$$\Delta E_h = (h\nu - E_g) - \Delta E_e \quad (3.2)$$

where ΔE_e is the energy difference between the conduction band and the initial energy of the photogenerated electron, and ΔE_h is the energy difference between the valence band and the photogenerated hole (see Fig. 3.1).

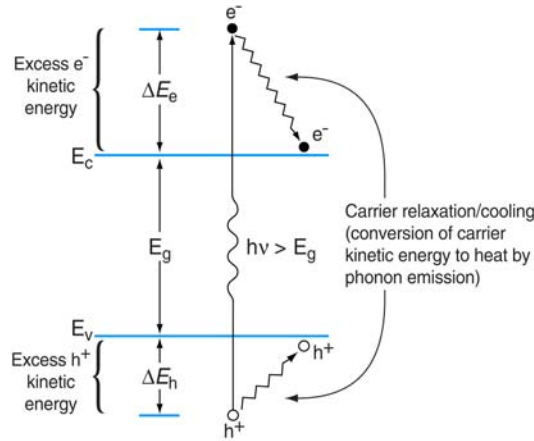


Figure 3.1 Hot-carrier relaxation/cooling in semiconductors. Source: Nozik (2001b).

In the Shockley–Queisser analysis, a major factor limiting the conversion efficiency to 31% is the loss of the absorbed photon energy above the semiconductor bandgap as heat through electron–phonon scattering and subsequent phonon emission, as the carriers relax to their respective band edges (bottom of the conduction band for electrons and top of the valence band for holes, as shown in Fig. 3.1) and equilibrate with the phonons. The present approach to reduce this loss and increase efficiency above the 31% limit has been to use a stack of cascaded multiple p – n junctions with descending bandgap values along the direction of the incident light such that higher energy photons are absorbed in the higher bandgap semiconductors and lower energy photons in the lower bandgap semiconductors [see Chapter 8 in Vol. I (Yamaguchi, 2001)]; this reduces the overall heat loss due to carrier relaxation via phonon emission. Such cells are generally called tandem solar cells. In the limit of an infinite

stack of bandgaps perfectly matched to the solar spectrum, the ultimate conversion efficiency at 1 Sun intensity increases to about 66%. For practical purposes, the stacks have been limited to two or three p - n junctions; actual efficiencies of about 32% have been reported in laboratory-scale photovoltaic cells with two cascaded p - n junctions, and a world-record 40.7% efficiency has recently been reported with three bandgaps, but with a solar concentration factor of 240 (press release, Spectrolab Inc., 6 Dec. 2006). Although the tandem cell efficiencies are very impressive, their cost is very high, and their main present use is for space applications. However, terrestrial applications using solar concentrators to reduce the net costs are being developed and may lead to competitive PV power costs (McConnell and Symko-Davies, 2006).

Other approaches to exceed the Shockley–Queisser limit have been under investigation for some time. In 1978, it was first proposed that conversion efficiencies above 31% in *single-bandgap* solar cells could be achieved by capturing the excess energy of electron–hole pairs created by the absorption of solar photons with energies larger than the bandgap to do useful work before these high-energy electron–hole pairs convert their excess kinetic energy (equal to the difference between the photogenerated electron energy and the conduction-band energy) to heat through phonon emission (Williams and Nozik, 1978; Boudreaux *et al.*, 1980; Ross and Nozik, 1982; Williams and Nozik, 1984; Nozik, 2001b). Subsequently, additional approaches to high efficiency have been proposed and studied. These approaches have been given the term ‘Third Generation Solar Cells’ by Martin Green (Green, 2001). Besides hot-carrier solar cells (Williams and Nozik, 1978; Boudreaux *et al.*, 1980; Ross and Nozik, 1982; Nozik, 2001b), they include solar cells that produce multiple electron–hole pairs per photon (Kolodinski *et al.*, 1993; Landsberg *et al.*, 1993; Green, 2001; Nozik, 2002; Schaller and Klimov, 2004), multiband and impurity-band solar cells (Luque and Martí, 1997; Green, 2001), thermophotovoltaic/thermophotonic cells (Green, 2001), and up-and-down conversion of the incident solar photons (Trupke *et al.*, 2002a and 2002b). In this chapter, only high-efficiency solar photon conversion approaches will be discussed that are based on the unique properties of nanosize semiconductor particles, interchangeably called nanocrystals (NCs) or quantum dots (QDs). These nanoscale particles exhibit unique and important quantisation effects that arise from the strong quantum confinement of electron and holes in the nanocrystals, which form the basis for several concepts for Third Generation Solar Photon Conversion devices. The diameters of NCs showing strong quantisation effects typically range from 1 nm to 20 nm, depending on the effective masses of the electrons and holes and their Bohr radii in the bulk semiconductor. The strongest confinement and quantisation occurs when the particle radius is less than the Bohr radius.

3.2 Quantisation effects in semiconductor nanostructures

When electrons and holes are confined by potential barriers to small regions of space where the dimensions of the confinement are less than the de Broglie wavelength of these charge carriers, pronounced quantisation effects develop; the length scale below which strong quantisation effects begin to occur ranges from about 5 nm to 25 nm for typical semiconductors (Groups IV, III–V, II–VI). In general, charge carriers in semiconductors can be confined by potential barriers in one, two, or three spatial dimensions. These regimes are termed quantum films, quantum wires, and quantum dots, respectively, and are shown in Fig. 3.2. Only quantum films and quantum dots will be discussed in this chapter.

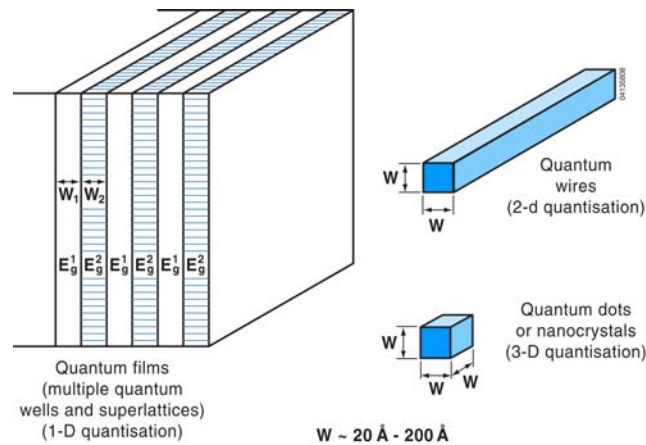


Figure 3.2 Three quantisation configurations depending on whether the confinement is in 1, 2, or 3 dimensions. Source: Nozik and Memming (1996).

Quantum films are more commonly referred to simply as quantum wells (QWs). This is because quantum films were the first quantised semiconductor nanostructures to be extensively studied, beginning in the 1970s (Dingle *et al.*, 1974). These structures were produced by molecular beam epitaxy (MBE) and metallo-organic chemical vapour deposition (MOCVD), and represented textbook examples of a *one-dimensional* quantum well (Dingle, 1987; Bastard, 1988; Jaros, 1989a; Weisbuch and Vinter, 1991; Yoffe, 1993). One-dimensional quantum wells are formed through epitaxial growth of alternating layers of semiconductor materials with different bandgaps. A single quantum well is formed from one semiconductor sandwiched between two layers of a second semiconductor having a larger bandgap; the centre

layer with the smaller bandgap semiconductor forms the QW, while the two layers sandwiching the centre layer create the potential barriers (Miller *et al.*, 1995), as shown in Fig. 3.3. Two potential wells are actually formed in the QW structure; one well is for conduction-band electrons, the other for valence-band holes. The well depth for electrons is the difference (*i.e.* the offset) between the conduction-band edges of the well and barrier semiconductors, while the well depth for holes is the corresponding valence-band offset. Multiple quantum well structures consist of a series of QWs (*i.e.* a series of alternating layers of wells and barriers). If the barrier thickness between adjacent wells is sufficient to prevent significant electronic coupling between the wells, then each well is electronically isolated; this type of structure is termed a *multiple quantum well* (MQW). On the other hand, if the barrier thickness is sufficiently thin to allow electronic coupling between wells (*i.e.* there is significant overlap of the electronic wavefunctions between wells), then the electronic charge distribution can become delocalised along the direction normal to the well layers. This coupling also leads to a broadening of the quantised electronic states of the wells; the new broadened and delocalised quantised states are termed *minibands* (see Fig. 3.4). A multiple QW structure that exhibits strong electronic coupling between the wells is termed a *superlattice* (SL). The critical barrier thickness at which miniband formation just begins to occur is about 40 Å (Dingle, 1987; Peterson *et al.*, 1988); the electronic coupling increases rapidly with decreasing thickness, and miniband formation is very strong below 20 Å (see Fig. 18, Chapter 1 of Dingle, 1987).

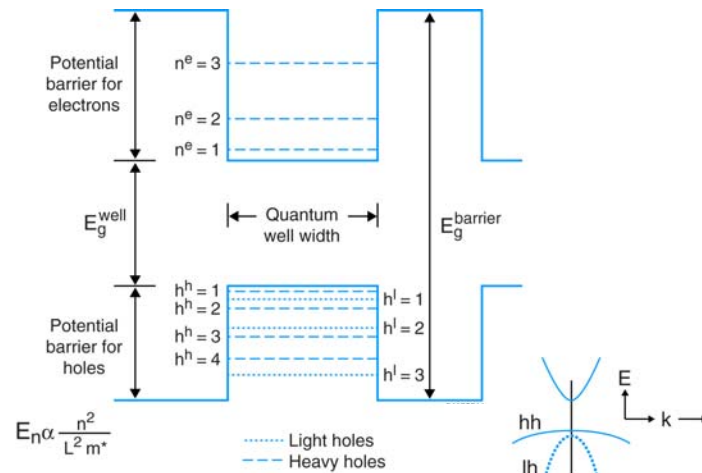


Figure 3.3 One-dimensionally confined quantum well created by a thin small-bandgap semiconductor layer sandwiched between two larger bandgap semiconductor layers. E_g^{well} is the bandgap of the bulk semiconductor, and E_g^{barrier} that of the barrier material.

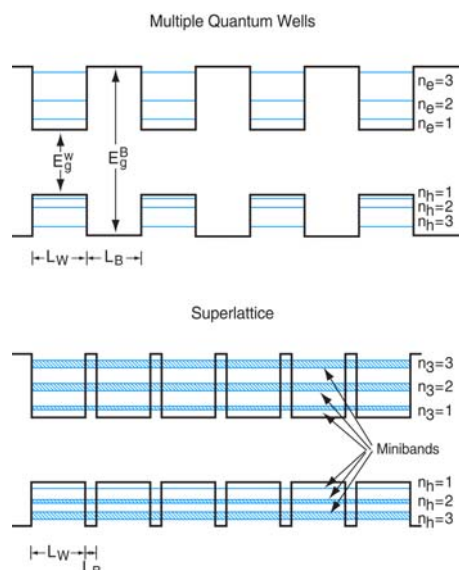


Figure 3.4 Energy levels of (a) multiple quantum wells and (b) superlattices. When the barriers are thick, the wells are isolated, there is no inter-well electronic coupling, and the quantised states are narrow. When the barriers are thin (<4 nm), inter-well electronic coupling occurs, the quantised states broaden, minibands form and electron delocalisation and transport can occur. Source: Nozik and Memming (1996).

Superlattice structures yield efficient charge transport normal to the layers, because the charge carriers can move through the minibands; the narrower the barrier, the wider the miniband and the higher the carrier mobility. Transport in MQWs with thick barriers requires thermionic emission of carriers over the barriers, or if electric fields are applied, field-assisted tunnelling through the barriers (Parsons *et al.*, 1990).

Quantum dots are nanosized semiconductor particles with confinement in all three dimensions. They are most commonly formed in two ways: through colloidal chemistry and via epitaxial deposition from the vapour phase.

3.2.1 Synthesis of semiconductor nanostructures

Quantum wells and superlattices

QWs are produced through epitaxial growth of crystalline films via either MBE or MOCVD. Both techniques are capable of creating epitaxial layers of sufficient quality to produce quantisation effects. These qualities include uniform film thickness and

interfacial abruptness (both within a few atomic layers), crystalline perfection and compositional uniformity.

Molecular beam epitaxy In the MBE technique, an ultrahigh-vacuum chamber is outfitted with a number of evaporation (*i.e.* effusion) cells, each controlled by a separate shutter, which supply fluxes of molecular beams of the desired atomic species. The beams can be turned on and off within 0.1 s. Growth rates are typically 5 Å/s. An ion-beam sputtering gun is used first to ion-etch the surfaces to remove impurities and imperfections. The substrate is maintained at ~500–700 C during growth. Spectroscopic capabilities such as mass spectrometry, Auger spectroscopy, and reflection high-energy electron diffraction are included in the chamber to control the process and provide data on film quality. Elemental species that can be produced by MBE include Ga, Al, In, As, Sb, Sn, Be, Ge, Se, Te, Cd, Hg, Zn, Mn, Pb and Si. The QW materials most commonly produced by MBE are the III–V semiconductor binary and ternary compounds, such as GaAs/Al_xGa_{1-x}As, GaSb/Al_xGa_{1-x}Sb, InAs/GaSb, Ga_xIn_{1-x}As/Al_xIn_{1-x}As, GaAs/Ga_xIn_{1-x}P₂ and InP/Ga_xIn_{1-x}As. In each case, the well material is the first, and the barrier material the second, compound. II–VI semiconductor QWs, such as CdTe/HgTe and ZnSe/Zn_{1-x}Mn_xS, have also been made.

Metallo-organic chemical vapour deposition MOCVD is primarily used to prepare III–V semiconductor QWs. In this process the Group III metals are introduced into a reaction chamber in the form of metallo-organic vapours that react at high temperatures with gaseous precursors of the non-metalloid Group V component of the desired semiconductor compound to form a crystalline, epitaxial film on a heated substrate. Ga, Al, and In are commonly introduced as trimethyl gallium (TMG), trimethyl aluminium (TMA) and trimethyl indium (TMI). As and P are usually introduced as arsine and phosphine, although less toxic compounds such as tertiary butyl arsine and tertiary butyl phosphine are also used. The organometallic compounds are kept in liquid form at a sufficiently low, but constant, temperature, and are swept into the reaction chamber at a controlled composition by sparging hydrogen gas at a controlled flow rate through the liquid metallo-organics. Gaseous arsine or phosphine is fed directly into the system from gas cylinders through flow and pressure controllers; ultra-pure hydrogen is used as the carrier gas for all reactant flows. The single-crystal substrate is placed on a graphite block that is heated either by RF or resistance heaters to 650–750 C. Growth rates are typically 5–10 Å/s. The reaction chamber is operated at either atmospheric or reduced pressure (50–100 torr). The low-pressure system can yield very sharp interfaces between the semiconductor heterojunctions and very uniform epilayers, as illustrated in Fig. 3.5.

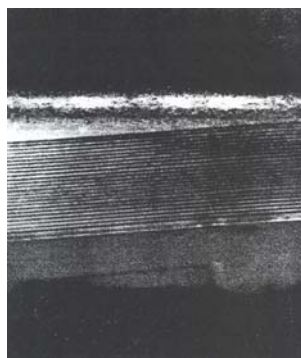


Figure 3.5 TEM of GaAs/Al_{0.32}Ga_{0.68}As superlattice structure. GaAs wells are 52 Å thick and the Al_{0.32}Ga_{0.68}As barriers are 17 Å thick. Source: Nozik (2001b).

Colloidal quantum dots

The most common approach to the synthesis of colloidal QDs is the controlled nucleation and growth of particles in a solution of chemical precursors containing the metal and the anion sources (controlled arrested precipitation) (Overbeek, 1982; Murray, 1995; Weller and Eychmüller, 1997). The technique of forming monodisperse colloids is very old, and can be traced back to the synthesis of gold colloids by Michael Faraday in 1857 (Overbeek, 1982; Murray, 1995). A common method for II–VI colloidal QD formation is to rapidly inject a solution of chemical reagents containing the Group II and Group VI species into a hot and vigorously stirred solvent containing molecules that can coordinate with the surface of the precipitated QD particles (Murray *et al.*, 1993; Murray, 1995; Weller and Eychmüller, 1997). A large number of nucleation centres are initially formed, and the coordinating ligands in the hot solvent prevent or limit particle growth via the normal process of Ostwald ripening (*i.e.* the growth of larger particles at the expense of smaller particles to minimise the higher surface free energy associated with smaller particles). Further improvement of the resulting size distribution of the QD particles can be achieved through size-selective precipitation (Murray *et al.*, 1993; Murray, 1995), whereby slow addition of a non-solvent to the colloidal solution of particles causes precipitation of the larger particles (the solubility of molecules with the same type of chemical structure decreases with increasing size). This process can be repeated several times to narrow the size distribution of II–VI colloidal QDs to a few percent of the mean diameter.

The synthesis of colloidal III–V QDs is more difficult than for II–VI QDs. Because the synthesis must be conducted in rigorously air-free and water-free atmospheres, it generally requires higher reaction temperatures and much longer reaction times, and it involves more complicated organometallic chemistry. For example, for the synthesis of InP QDs (Mićić *et al.*, 1994, 1995, 1996 and 1997; Guzelian *et al.*, 1996b; Banin *et al.*, 1997) an indium salt [$\text{In}(\text{C}_2\text{O}_4)_3$, InF_3 , or InCl_3] is reacted with trimethylsilylphosphine [$\text{P}(\text{Si}(\text{CH}_3)_3)_3$] in a solution of trioctylphosphine oxide (TOPO) and trioctylphosphine (TOP) to form a soluble InP organometallic precursor species that contains In and P in a 1:1 ratio. The InP precursor species in the TOPO/TOP solution is then heated for several days at a temperature ranging from 270 C to 290 C, depending on the desired QD properties. One difference with the synthesis of II–VI materials is that more than one day of heating at reaction temperature is required to form crystalline III–V QDs, whereas II–VI QDs form immediately on injection of the reactants into the hot TOPO/TOP solution.

The resulting InP QDs contain a capping layer of TOPO, which can be readily exchanged for several other types of capping agents, such as thiols, pyridines, amines and polymers. The size distribution of the InP QDs can be further narrowed to less than 10% through selective precipitation techniques. Finally, they can be studied in the form of colloidal solutions, powders, or dispersed in transparent polymers or organic glasses (for low-temperature studies); capped InP QDs recovered as powders can also be redissolved to form transparent colloidal solutions.

In addition to InP QDs, similar methods have been used to produce III–V QDs of GaP (Mićić *et al.*, 1995; Mićić and Nozik, 1996), GaInP_2 (Mićić *et al.*, 1995; Mićić and Nozik, 1996), GaAs (Olshavsky *et al.*, 1990; Uchida *et al.*, 1992; Nozik *et al.*, 1993; Mićić *et al.*, 1995; Mićić and Nozik, 1996), and InAs (Guzelian *et al.*, 1996a); higher temperatures (400 C) are required for GaP and GaInP_2 QDs.

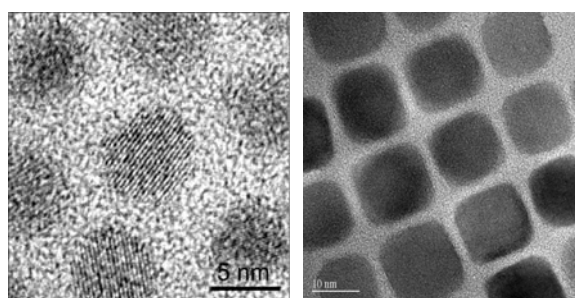


Figure 3.6 Transmission electron micrograph of PbSe QD spheres (5 nm dia) and cubes (10 nm). The spherical QD dots contain about 20 rows of atoms, or ~6000 atoms.

Group IV–VI QDs can also be readily formed and the size and shape easily controlled [(Murphy *et al.*, 2006) and references therein]. Figure 3.6 shows TEM images of PbSe QD spheres and cubes.

Stranski–Krastanov epitaxial growth

QDs can also be produced via epitaxial growth from the vapour phase on appropriate substrates; the growth can be effected using either MBE or MOCVD. In the Stranski–Krastanov (SK) mode of QD formation, a thin semiconductor film is deposited by MBE or MOCVD onto another semiconductor substrate material that has a mismatch in lattice constant with the material being deposited. If the lattice mismatch is sufficiently high and the film has a higher surface energy than the substrate, then after the initial growth of a few monolayers (called the wetting layer), subsequent deposition forms small islands which are in the QD size regime. The size, shape, perfection and density of these SK QDs can be controlled by controlling the conditions in the MBE or MOCVD reactor. The bulk of work on these SK QDs has been on $\text{InGa}_x\text{As}_{1-x}$ QDs grown on GaAs substrates (Sugawara, 1999); InP QDs grown on GaAs or $\text{Al}_x\text{Ga}_{1-x}\text{As}$ substrates have also been reported (Hanna *et al.*, 1997). The size of SK QDs is generally larger than that of colloidal QDs; they typically have a lateral dimension of 1000–2000 Å and a height of 100–500 Å. For III–V semiconductors these sizes produce quantisation effects because their Bohr radius is large, but the degree of quantisation is not as strong as that in colloidal QDs where the sizes range from 15 Å to 100 Å. The shape of the InAs/GaAs SK QDs has been reported to be a square-based pyramid (Ruminov and Scheerschmidt, 1995); however, other results suggest a parallelogram base and C_{2v} symmetry (Lee *et al.*, 1998; Yang *et al.*, 2000).

Another configuration of SK type QDs is possible. In this case, a QW which has a thin outer barrier (about 100 Å) is first formed. SK islands of another semiconductor material having a lattice mismatch with the barrier are then deposited on top of the barrier. These SK islands produce a parabolic strain field which propagates down through the barrier into the QW. In the case of InP SK islands (called stressor islands) formed on an AlGaAs/GaAs/AlGaAs QW, the strain field expands the GaAs lattice and reduces the GaAs bandgap just below the InP islands (Hanna *et al.*, 1997). Thus, the GaAs QW beneath the InP stressor islands is converted into a QD since the one-dimensional confinement of carriers is transformed into three-dimensional confinement. An important feature of this type of QD is that both the well and barrier region of the QD are made of the same material. This eliminates interface defects and interface states that complicate the relaxation behaviour of photogenerated carriers.

3.2.2 Energy levels in quantum wells, superlattices and quantum dots

Quantum wells and superlattices

The confinement of electrons or holes in potential wells leads to the creation of discrete energy levels in the wells, compared with the continuum of states in bulk material; quantisation also leads to a major change in the density of states. The energy levels can be calculated by solving the Schrödinger equation for the well-known particle in a box problem. Using the effective mass envelope function approximation (Bastard, 1981 and 1982; Altarelli, 1985; Bastard and Brum, 1986), the electron wavefunction ψ_n is then

$$\psi_n = \sum_{W,B} e^{ik \cdot r} U_k^{W,B}(r) \phi_n(z) \quad (3.3)$$

where W and B refer to well and barrier, respectively; k is the transverse electron wave vector; z is the growth direction; $U_k^{W,B}(r)$ is the Bloch wavefunction in W or B ; and $\phi_n(z)$ is the envelope wavefunction. The envelope wavefunction is determined by solving the Schrödinger equation

$$\left(\frac{\hbar^2}{2m^*} \frac{\partial^2}{\partial z^2} + V_c(z) \right) \phi_n(z) = E_n \phi_n(z) \quad (3.4)$$

where $V_c(z)$ is the potential barrier function, and E_n the quantised energy level in the well.

Considering first isolated QWs that are infinitely deep, the solution to eq. 3.4 becomes very simple since the wavefunctions must be zero at the well-barrier interfaces. This leads to the well-known solutions

$$\psi_n = A \sin \frac{n\pi z}{L_w} \quad (3.5)$$

$$E_n = \frac{\hbar^2}{2m^*} \left(\frac{n\pi}{L_w} \right)^2, \quad n=1, 2, 3, \dots \quad (3.6)$$

where L_w is the well width.

For the more realistic case of finite barrier heights, eq. 3.4 cannot be solved exactly, but requires either a graphical or numerical solution. For the case where the effective masses in the well and barrier are taken to be equal

$$\phi_n(z) = A \cos kz \text{ (inside well)} \quad (3.7)$$

$$\phi_n(z) = B \exp[-k(z - L_w/2)] \text{ (outside well)} \quad (3.8)$$

where

$$E_n = \frac{\hbar^2 k^2}{2m^*} - V_o, \quad -V_o < E_n < 0 \quad (3.9)$$

and

$$\cos(kL_w/2) = k/k_o \text{ for } \tan kL_w/2 > 0 \quad (3.10)$$

$$\sin(kL_w/2) = k/k_o \text{ for } \tan kL_w/2 > 0 \quad (3.11)$$

where

$$k_2 = 2m^* V_o / \hbar^2 \quad (3.12)$$

and V_o is the barrier height.

There is always at least one state in the well no matter how thin it is; the number of bound states is given by

$$1 + \text{Int} \left[\frac{(2m^* V_o L_w^2)^{1/2}}{\pi^2 \hbar^2} \right] \quad (3.13)$$

where $\text{Int}[x]$ indicates the integer part of the argument.

The calculation of the hole levels is much more complicated since the band structure of many important semiconductors has hole bands with fourfold degeneracy at $k=0$. This leads to heavy and light holes with different effective masses. Consequently, a double set of hole energy levels is formed in the QW with different spacings between levels—one set for the light holes, the second set for the heavy holes, as shown in Fig. 3.3. Solutions to the problem have been reported for both infinite (Bastard, 1981; Altarelli, 1985) and finite (Bastard and Brum, 1986) potential barriers.

For superlattices, several approaches have been used to calculate the energy level structure of the minibands (Bastard, 1981; Altarelli, 1985; Bastard and Brum, 1986). One approach is to use a tight-binding model for the multiple wells, leading to a Bloch-like envelope function of the form (Dingle *et al.*, 1975)

$$\psi_q^i(z) = \frac{1}{\sqrt{N}} \sum_n e^{ipnx} \phi^i(z - nx) \quad (3.14)$$

where $\phi^i(z - nx)$ is the i th wavefunction of the QW and q is the Bloch wave vector.

This approach leads to results that are summarised in Fig. 18, Chapter 1 of Dingle (1987), which shows the energy levels as a function of barrier thickness. That figure shows the formation of minibands with energy bandwidths that increase rapidly with decreasing barrier widths below 40 Å.

The density of states (DOS) also shows profound changes with the dimensionality of quantisation. For ideal bulk semiconductors with simple parabolic energy bands, the DOS has a square-root dependence on electron energy

$$N(E)_{\text{bulk}} = \sqrt{2} (m^*)^{3/2} \pi^2 \hbar^{-3} E^{1/2} \quad (3.15)$$

For an ideal QW, the DOS shows a step-like function with each plateau having a DOS of

$$N(E)_{\text{QW}} = nm^* / \pi \hbar^2 \quad (3.16)$$

where n is the quantum number of the state.

For a superlattice, the steepness of the steps is destroyed by the dispersion of the N states in a miniband (Dingle, 1987; Weisbuch and Vinter, 1991)

$$N(E)_{\text{SL}} = N \frac{m^*}{\pi \hbar^2} \arccos \left(\frac{\varepsilon_i - E_i - S_i}{2 x_i} \right) \quad (3.17)$$

where ε_i is the energy dispersion caused by superlattice formation and the i th quantum state, E_i is the energy of the i th quantum state for the isolated quantum well, S_i is the electron transfer matrix describing transmission through the potential barrier from the i th quantum state, and x_i is the square root of the electron transmission probability at the i th quantum state.

Figure 3.7 shows these DOS functions for bulk semiconductors, QWs, superlattices, quantum wires and quantum dots; the DOS for quantum dots exhibits discrete values at the discrete quantised energy levels.

Quantum dots

The earliest and simplest treatment of the electronic states of a QD is based on the effective mass approximation (EMA); the simple EMA treatment can be improved by incorporating the $\mathbf{k} \bullet \mathbf{p}$ approach, which has commonly been used to calculate the electronic structure of bulk semiconductor and QW structures.

The EMA rests on the assumption that if the QD is larger than the lattice constants of the crystal structure, then it will retain the lattice properties of the infinite crystal and the same values of the carrier effective masses; the electronic properties of the

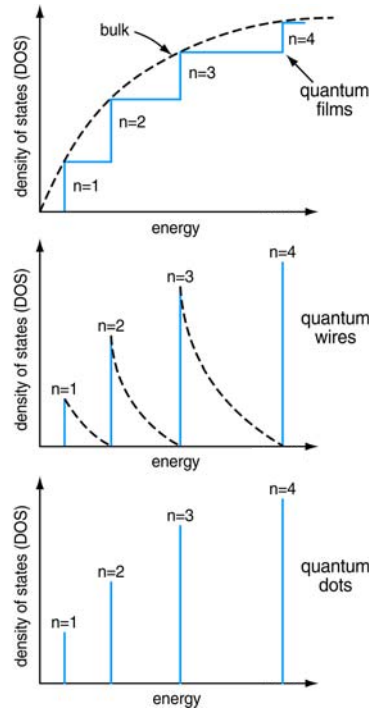


Figure 3.7 Density of states for bulk semiconductors, quantum films, quantum wires and quantum dots.
Source: Nozik and Memming (1996).

QD can then be determined by simply considering the modification of the energy of the charge carriers produced by the quantum confinement. Thus, the electronic properties are determined by solving the Schrödinger equation for a particle in a 3-dimensional box. The zeroth order approximation is a perfectly spherical QD with infinite potential walls at the surface. Strong confinement is defined as the case where the QD size is small compared with the de Broglie wavelength of electrons in the box or small compared with the Bohr radius of the bound electron-hole pair (exciton) in the bulk; this is the case for II-VI and III-V semiconductors. Taking into account the Coulomb interaction between electrons and holes, which is enhanced due to confinement in the QD, the Hamiltonian can then be written

$$\hat{H} = -\frac{\hbar^2 \nabla_e^2}{2m_e^*} - \frac{\hbar^2 \nabla_h^2}{2m_h^*} - \frac{q^2}{\epsilon |r_e - r_h|} + V_e(r_e) + V_h(r_h) \quad (3.18)$$

and

$$\hat{H}\psi(r) = E\psi(r) \quad (3.19)$$

where V_e and V_h are the confining potentials, r_e and r_h are the distances of the electron and hole from the centre of the QD, and ϵ is the permittivity of the semiconductor.

Analytical solutions of eqs. 3.18 and 3.19 are difficult because centre-of-mass motion and reduced mass motion cannot be separated as independent coordinates. Various approaches to solving this problem have been used. These include perturbation theory (Brus, 1984 and 1986), variational calculations (Schmidt and Weller, 1986; Kayanuma, 1988; Ekimov *et al.*, 1989; Kayanuma and Momiji, 1990; Takagahara, 1993a and 1993b; Kagan, 1996, p. 287), matrix diagonalisation (Hu *et al.*, 1990; Park *et al.*, 1990), and Monte Carlo methods (Pollock and Koch, 1991). Perturbation theory (Brus, 1984 and 1986) and variational calculations (Kayanuma, 1988) lead to a solution of the form

$$E_{\min} = \frac{\hbar^2 \pi^2}{2R^2} \left[\frac{1}{m_e^*} + \frac{1}{m_h^*} \right] - \frac{1.8q^2}{\epsilon_s R} - 0.25E_{\text{Ryd}}^* \quad (3.20)$$

where $E_{\min}$ is the lowest energy separation between hole and electron states in the QD, E_{Ryd}^* is the bulk exciton binding energy in meV, and R is the QD radius. $E_{\min}$ is often referred to as the bandgap of the QD since it represents the threshold energy for photon absorption, blue-shifted from the bulk bandgap, E_g .

Improvements in the EMA model involve accounting for band non-parabolicity and hole-state mixing (Grigoryan *et al.*, 1990; Sercel and Vahala, 1990; Nomura and Kobayashi, 1991; Efros, 1992; Koch *et al.*, 1992; Ekimov *et al.*, 1993), finite barrier heights (Kayanuma and Momiji, 1990), and surface polarisation (Banyai *et al.*, 1992).

In the effective mass $\mathbf{k} \bullet \mathbf{p}$ approach, the QD wavefunctions are expanded as a linear combination of N_b bulk Bloch functions at the $\mathbf{k} = 0$ Brillouin zone centre (Γ point) (Grigoryan *et al.*, 1990; Sercel and Vahala, 1990); for these calculations, values of N_b up to 8 (termed the 8-band $\mathbf{k} \bullet \mathbf{p}$ model) have been used (Banin *et al.*, 1998). The effective mass $\mathbf{k} \bullet \mathbf{p}$ approach has been very extensively used for QDs, but its rigour has been debated (Fu *et al.*, 1997, 1998a and 1998b; Efros and Rosen, 1998 and 2000). An approach based on direct diagonalisation and pseudopotentials is proposed as a better theoretical basis on which to calculate the electronic states in QDs (Fu *et al.*, 1997 and 1998a). A comparison of the calculated electron and hole states for a 28 Å InP QD using a 6-band $\mathbf{k} \bullet \mathbf{p}$ model (Fu and Zunger, 1998) and the direct diagonalisation pseudopotential model (Fu and Zunger, 1998) indicates a higher density of electronic states for the latter than the former.

Because of the three-dimensional spatial confinement in QDs, the solution of the Schrödinger equation results in describing the electronic states in the QD by three quantum numbers plus spin. A commonly used notation (Woggon, 1997; Xia, 1989) labels the electron states as nL_e and the hole states as nL_h , where n is the principal quantum number (1,2,3 *etc.*), L is the orbital angular momentum (S, P, D *etc.*), and F is the total angular momentum ($F = L + J$, and $J = L + S$), where S is the spin, and the projection of F along a magnetic axis is $m_F = -F$ to $+F$. Thus electron states become $1S_e, 2S_e, 1P_e$ *etc.*, and hole states become $1S_{1/2}, 1S_{3/2}, 1P_{1/2}$ *etc.* For optical transitions in ideal spherical QDs, the selection rules are $\Delta n = 0$, $\Delta L = 0, \pm 2$; and $\Delta F = 0, \pm 1$. These rules can be broken by non-spherical QDs and strong hole-state mixing.

3.3 Optical spectroscopy of quantum wells, superlattices and quantum dots

3.3.1 Quantum wells and superlattices

Optical transitions between quantum levels can occur on excitation with light. The interband transition probability is the product of an optical matrix element (M) times a DOS term. M involves the electric dipole operator and can be expressed as (Weisbuch and Vinter, 1991)

$$M = \left\langle \phi_e(z) e^{ik_e \cdot r} U_{cke}(r) \middle| \hat{\mathbf{n}} \middle| \phi_h(z) e^{ik_h \cdot r} U_{vkh}(r) \right\rangle \quad (3.21)$$

where $\phi_e(z)$ and $\phi_h(z)$ are the electron and hole envelope wavefunctions, k_e and k_h are electron and hole wave vectors, $\hat{\mathbf{n}}$ is the polarisation vector of the light, and $U_{cke}(r)$ and $U_{vkh}(r)$ are the Bloch functions. The integration of eq. 3.21 produces a factor $[\phi_e(z)\phi_h(z)]$ which becomes unity for transitions between electron and hole states with the same quantum number. Thus the selection rule for optical interband transitions is that $\Delta n = 0$. Some additional factors affecting optical transitions are: (a) heavy hole–electron transitions are expected to be three times stronger than light hole–electron transitions; (b) transitions involving different n values become possible in the presence of an electric field and also when finite wells are considered; (c) parity is a good quantum number, and transitions involving the same parity are allowed.

In light of the above discussion, the optical absorption spectrum of a quantum film is expected to consist of a series of steps, with the position of these steps corresponding to the transitions between heavy or light hole quantum states and electron quantum states following the selection rule $\Delta n = 0$. Furthermore, since the widths of the wells are commonly smaller than the calculated diameter of an exciton, the exciton binding

energy is greatly increased in QWs, and the excitons can be stable even at room temperature. Thus the absorption spectra of QWs can be expected to show exciton peaks, even at room temperature, which occur at energies below the step. Such spectra have indeed been observed by many workers (Dingle *et al.*, 1974 and 1975).

3.3.2 Quantum dots

The optical absorption and emission properties of quantum dots follow the same general photophysics as QWs and SLs in that the probability for dipole-allowed optical transitions is proportional to the matrix element connecting the initial [$\psi_i(r)$] and final [$\psi_f(r)$] states through the dipole operator ($\hat{\mathbf{n}}$)

$$M_{\text{qd}} = \langle \psi_f(r) | \hat{\mathbf{n}} | \psi_i(r) \rangle \quad (3.22)$$

The wavefunctions for three-dimensional confinement have three quantum numbers (n, l, m) plus spin. The selection rules for dipole-allowed absorption and emission were given above for interband transitions. For intraband transitions between the ladder of electron or hole states, the selection rules for the simplest case of non-interacting electrons and holes are $\Delta n \neq 0$; $\Delta L = 0, \pm 1$; $\Delta m = 0, \pm 1$.

As an example, Fig. 3.8 shows the absorption and emission spectra of initially prepared InP QDs with a mean diameter of 32 Å. Samples were prepared by dispersing washed QDs in toluene, and optical absorption and photoluminescence spectra collected at room temperature. Other III–V and II–VI QDs exhibit similar behaviour. The absorption spectrum shows an excitonic peak at about 590 nm. The photoluminescence (PL) spectrum (excitation at 500 nm) shows two emission bands: a weaker one near the band edge with a peak at 655 nm, and a second, stronger, broader band that peaks above 850 nm. The PL band with deep red-shifted subgap-emission peaking above 850 nm is attributed to radiative recombination in QD surface states produced by phosphorus vacancies (Mićić *et al.*, 1996). These surface defects and the accompanying red-shifted emission can be completely removed by etching the InP QDs with HF; intense band-edge emission from the QDs can then be observed. After etching, the PL intensity increases by a factor of ten, and the near-infrared emission produced by the deep surface traps is completely removed (Mićić *et al.*, 1996). The near-band-edge quantum yield (QY) increases from 30% to 60% as the temperature decreases from 300 K to 10 K (Mićić *et al.*, 1996) (see Fig. 3.9).

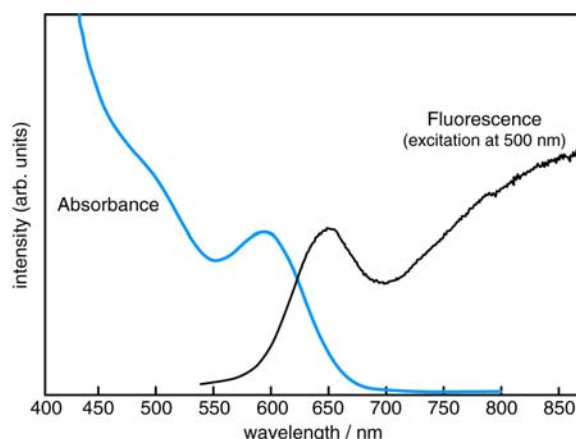


Figure 3.8 Absorption and emission spectra at 300 K for InP QD with excess In^{3+} (In:P = 1.6:1).
Source: Mičić *et al.* (1996).

The absorption spectrum in Fig. 3.8 shows an excitonic peak at about 590 nm. The photoluminescence (PL) spectrum (excitation at 500 nm) shows two emission bands: a weaker one near the band edge with a peak at 655 nm, and a second, stronger, broader band that peaks above 850 nm. The PL band with deep red-shifted subgap-emission peaking above 850 nm is attributed to radiative recombination in QD surface states produced by phosphorus vacancies (Mičić *et al.*, 1996). These surface defects and the accompanying red-shifted emission can be completely removed by etching the InP QDs with HF; intense band-edge emission from the QDs can then be observed. After etching, the PL intensity increases by a factor of ten, and the near-infrared emission produced by the deep surface traps is completely removed (Mičić *et al.*, 1996). The near-band-edge quantum yield (QY) increases from 30% to 60% as the temperature decreases from 300 K to 10 K (Mičić *et al.*, 1996) (see Fig. 3.9).

Figure 3.9 shows the room-temperature absorption spectra as a function of QD size ranging from 26 Å to 60 Å, measured by transmission electron microscopy (TEM) (Mičić *et al.*, 1997). The absorption spectra show one or more broad excitonic peaks which reflect substantial inhomogeneous line broadening arising from the QD size distribution; as expected, the spectra shift to higher energy as the QD size decreases (Mičić *et al.*, 1997). The colour of the InP QD samples changes from deep red (1.7 eV) to green (2.4 eV) as the diameter decreases from 60 Å to 26 Å. Bulk InP is black with a room-temperature bandgap of 1.35 eV and an absorption onset at 918 nm. Higher energy transitions above the first excitonic peak in the absorption spectra can also be easily seen in QD samples with mean diameters equal to or greater

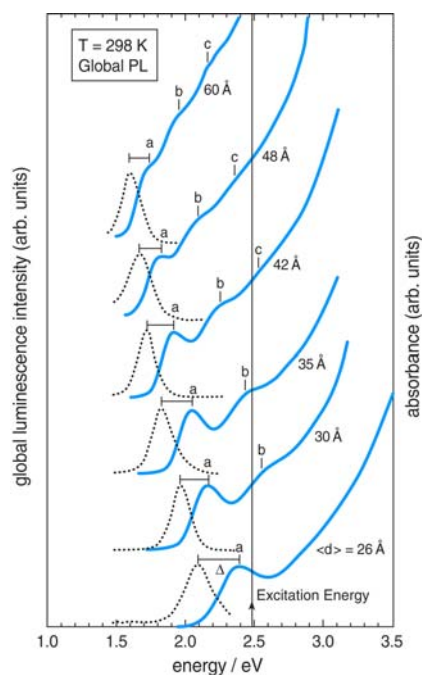


Figure 3.9 Absorption (solid line) and global photoluminescence (dotted line) spectra at 298 K for colloidal ensembles of InP quantum dots with different mean diameters. All samples were photoexcited at 2.48 eV. Source: Mićić *et al.* (1997).

than 30 Å. The spread in QD diameters is generally about 10% and is somewhat narrower in samples with larger mean diameters; this is why higher energy transitions can be resolved for the larger-sized QD ensembles. All of the prepared QD nanocrystallites are in the strong confinement regime since the Bohr radius of bulk InP is ~ 100 Å. Figure 3.9 shows typical room-temperature absorption and global emission spectra of InP colloids with mean particle diameters ranging from 26 Å to 60 Å as measured by TEM (Mićić *et al.*, 1997); the excitation energy for all the QD sample ensembles in Fig. 3.9 was 2.48 eV, well above their absorption onset in each case.

The global emission peaks in Fig. 3.9 show an increasing red shift from the first excitonic absorption peak as the mean QD size decreases from 60 Å to 26 Å. The global (‘nonresonant’) red shift is as large as 300 meV for samples with the smallest mean diameter (Mićić *et al.*, 1997). Global PL is defined as that observed when the excitation energy is much higher than the energy of the absorption threshold exhibited in the absorption spectrum produced by the ensemble of QDs in the sample; thus a large fraction of all the QDs in the sample is excited. The resulting global PL shows a

very broad peak (line width of 175–225 meV) that is red-shifted by 100–300 meV from the first absorption peak. The broad PL line width is caused by inhomogeneous line broadening arising from the ~10% size distribution sampled in the global PL experiment. The large global red shift is caused by the volume dominance of the larger particles in the size distribution; the latter will absorb most of the incident photons and will show large red shifts since the PL excitation energy is well above their lowest transition energy.

3.4 Hot carrier cooling dynamics in quantum-confined semiconductors

As indicated in Section 3.1, the maximum solar photon conversion efficiency for a single-gap absorber is limited to 31% if the photogenerated hot electrons and holes created with energies above the conduction and valence bands convert their initial excess kinetic energy into heat through phonon emission. If the hot electrons and holes can convert their excess energy into electrical or chemical free energy instead of heat, then the ultimate thermodynamic efficiency can be increased to 66% (Ross and Nozik, 1982). To do this, the rate of conversion to useful free energy must be faster than the rate of cooling via phonon emission. It is therefore important to determine and understand the dynamics of hot electron and hole cooling, and how these dynamics can be modified and controlled in quantum-confined structures; this is the subject of this section.

3.4.1 Quantum wells and superlattices

In this section we will discuss the electron and hole cooling dynamics in III–V QWs and SLs compared with bulk III–V semiconductors. These dynamics can be determined from several types of time-resolved PL experiments. One technique involves hot luminescence non-linear correlation (Xu and Tang, 1984; Rosker *et al.*, 1986; Edelstein *et al.*, 1987), which is a symmetrised pump–probe type of experiment. Figure 2 of Edelstein *et al.* (1987) compares the hot-electron relaxation times as a function of the electron energy level in the well for bulk GaAs and a 20-period MQW of GaAs/Al_{0.38}Ga_{0.62}As containing 250 Å GaAs wells and 250 Å Al_{0.38}Ga_{0.62}As barriers. For bulk GaAs, the hot-electron relaxation time varies from about 5 ps near the top of the well to 35 ps near the bottom of the well. For the MQW, the corresponding hot-electron relaxation times are 40 ps and 350 ps.

Another method uses time-correlated single-photon counting to measure PL lifetimes of hot electrons. Figure 3.10 shows three-dimensional plots of PL intensity as a function of energy and time for both bulk GaAs and a 250 Å/250 Å GaAs/Al_{0.38}Ga_{0.62}As MQW (Rosenwaks *et al.*, 1993). It is clear from these plots that the MQW sample exhibits much longer-lived hot luminescence (*i.e.* luminescence above the lowest $n = 1$ electron to heavy-hole transition at 1.565 eV) than bulk GaAs. Depending on the emitted photon energy, the hot PL for the MQW is seen to exist beyond times ranging from hundreds to several thousand picoseconds. On the other hand, the hot PL intensity above the bandgap (1.514 eV) for bulk GaAs is negligible over most of the plot; it is only seen at the very earliest times and at relatively low photon energies.

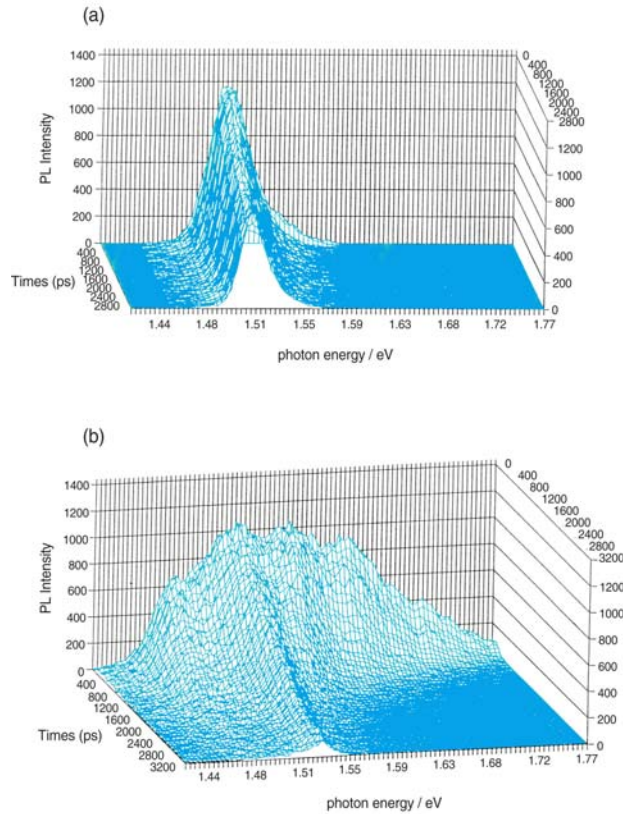


Figure 3.10 Three-dimensional plots of photoluminescence intensity *versus* time and photon energy for (a) bulk GaAs and (b) 250 Å GaAs/250 Å Al_{0.38}Ga_{0.62}As MQW. Source: Rosenwaks *et al.* (1993).

Rosenwaks *et al.* (1993) performed calculations on the PL intensity *versus* time and energy data to determine the time dependence of the quasi-Fermi level, electron temperature, electronic specific heat, and ultimately the dependence of the characteristic hot-electron cooling time on electron temperature.

The cooling, or energy-loss, rate for hot electrons is determined by LO phonon emission through electron–LO-phonon interactions. The time constant characterising this process can be described by the following expression (Ryan *et al.*, 1984; Cai *et al.*, 1986; Christen and Bimberg, 1990)

$$P_e = -\frac{dE}{dt} = \frac{\hbar\omega_{LO}}{\tau_{avg}} \exp(-\hbar\omega_{LO}/kT_e) \quad (3.23)$$

where P_e is the power loss of electrons (*i.e.* the energy-loss rate), $\hbar\omega_{LO}$ is the LO phonon energy (36 meV in GaAs), T_e is the electron temperature, and τ_{avg} is the time constant characterising the energy-loss rate.

The electron energy-loss rate is related to the electron-temperature decay rate through the electronic specific heat. Since the electron distribution becomes degenerate at high light irradiance, the classical specific heat is no longer valid. Hence the temperature- and density-dependent specific heat for both the QW and bulk samples need to be calculated as a function of time in each experiment so that τ_{avg} can be determined.

The results of such calculations, presented in Fig. 2 of Rosenwaks *et al.* (1993), show a plot of τ_{avg} *versus* electron temperature for bulk and MQW GaAs at high and low carrier densities. At a high carrier density [$n \sim (2-4) \times 10^{18} \text{ cm}^{-3}$], the τ_{avg} values for the MQW are much higher ($\tau_{avg} = 350-550 \text{ ps}$ for T_e between 440 K and 400 K) compared with bulk GaAs ($\tau_{avg} = 10-15 \text{ ps}$ over the same T_e interval). On the other hand, at a low carrier density [$n \sim (3-5) \times 10^{17} \text{ cm}^{-3}$] the differences between the τ_{avg} values for bulk and MQW GaAs are much smaller.

A third technique to measure cooling dynamics is PL upconversion. For example, Rosenwaks *et al.* (1993) recorded time-resolved luminescence spectra at room temperature for a 4000-Å bulk GaAs sample at incident pump powers of 25, 12.5 and 5 mW. They determined electron temperatures by fitting the high-energy tails of the spectra, choosing only the region which is linear on a semilogarithmic plot for the fit. The carrier densities for the sample were 1×10^{19} , 5×10^{18} , and $2 \times 10^{18} \text{ cm}^{-3}$, corresponding to the incident excitation powers of 25, 12.5 and 5 mW, respectively. Spectra for the MQW sample were recorded at the same pump powers as the bulk. Figure 3.11 shows τ_{avg} for bulk and MQW GaAs at the three irradiances, again showing the much slower cooling in MQWs (by up to two orders of magnitude).

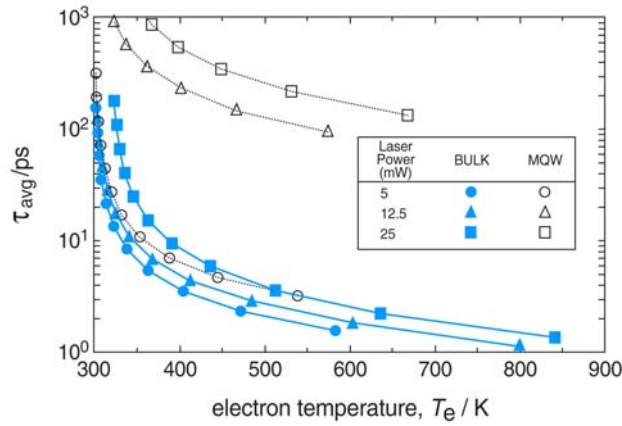


Figure 3.11 Time constant for hot-electron cooling (τ_{avg}) versus electron temperature for bulk GaAs and GaAs multiple quantum wells at three excitation intensities. Source: Rosenwaks *et al.* (1993).

The difference in hot-electron relaxation rates between bulk and quantised GaAs structures is also reflected in time-integrated PL spectra. Typical results are presented [see Fig. 7 in Nozik (2001b)] for single-photon counting data taken with 13 ps pulses of 600 nm light at 800 kHz focussed to about 100 μm with an average power of 25 mW (Nozik *et al.*, 1990). The time-averaged electron temperatures obtained from fitting the tails of these PL spectra to the Boltzmann function show that the electron temperature varies from 860 K for the 250 Å/250 Å MQW to 650 K for the 250 Å/17 Å superlattice, while bulk GaAs has an electron temperature of 94 K, which is close to the lattice temperature (77 K). The variation in the electron temperatures between the quantised structures can be attributed to differences in electron delocalisation between MQWs and SLs, and the associated non-radiative quenching of hot-electron emission.

As shown above, the hot-carrier cooling rates depend on photogenerated carrier density; the higher the electron density, the slower the cooling rate. This effect is also found for bulk GaAs, but it is much weaker than in quantised GaAs. The most generally accepted mechanism for the decreased cooling rates in GaAs QWs is an enhanced hot phonon bottleneck (Lugli and Goodnick, 1987; Joshi and Ferry, 1989; Campos *et al.*, 1992). In this mechanism, a large population of hot carriers produces a non-equilibrium distribution of phonons (in particular, optical phonons, which are the type involved in the electron-phonon interactions at high carrier energies), because the optical phonons cannot equilibrate fast enough with the crystal bath; these hot phonons can be reabsorbed by the electron plasma to keep it hot (see Fig. 3.12). In

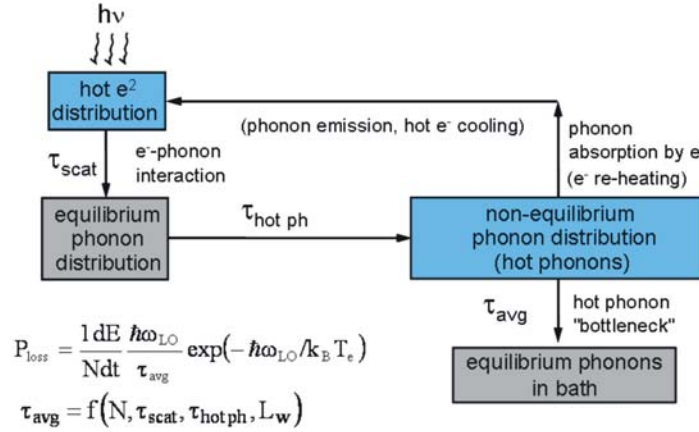


Figure 3.12 Hot phonon bottleneck to slow hot-electron cooling in QWs. At high light intensity, hot electrons produce hot phonons which can reheat electrons via phonon absorption to keep them hot. P_{loss} is power loss per electron.

QWs, the phonons are confined in the well and they exhibit slab modes (Campos *et al.*, 1992), which enhance the hot phonon bottleneck effect.

The first prediction of slowed cooling at low light intensities in quantised structures was made by Boudreaux, Williams, and Nozik (Boudreaux *et al.*, 1980). They anticipated that cooling of carriers would require multiphonon processes when the quantised levels are separated in energy by more than phonon energies. They analysed the expected slowed cooling time for hot holes at the surface of highly doped *n*-type TiO_2 semiconductors, where quantised energy levels arise because of the narrow space-charge layer (*i.e.* depletion layer) produced by the high doping level. The carrier confinement in this case is produced by the band bending at the surface; for a doping level of $1 \times 10^{19} \text{ cm}^{-3}$, the potential well can be approximated as a triangular well extending 200 Å from the semiconductor bulk to the surface and with a depth of 1 eV at the surface barrier. The multiphonon relaxation time was estimated from the approximation

$$\tau_c \sim \omega^{-1} \exp(\Delta E / kT) \quad (3.24)$$

where τ_c is the hot-carrier cooling time, ω is the phonon frequency, and ΔE is the energy separation between quantised levels. For strongly quantised electron levels, with $\Delta E > 0.2 \text{ eV}$, τ_c could be $>100 \text{ ps}$ according to eq. 3.24.

However, carriers in the space-charge layer at the surface of a heavily doped semiconductor are only confined in one dimension, as in a quantum film. This

quantisation regime leads to discrete energy states which have dispersion in $\mathbf{k}$ -space (Jaros, 1989b). This means the hot carriers can cool by undergoing interstate transitions which require only one emitted phonon followed by a cascade of single phonon intrastate transitions; the bottom of each quantum state is reached by intrastate relaxation before an interstate transition occurs. Thus the simultaneous and slow multiphonon relaxation pathway can be bypassed by single-phonon events, and the cooling rate increases correspondingly.

3.4.2 Quantum dots

As previously discussed, slowed hot-electron cooling in QWs and SLs that is produced by a *hot* phonon bottleneck requires very high light intensities in order to create the required photogenerated carrier density of greater than about $1 \times 10^{18} \text{ cm}^{-3}$. This required intensity, possible with laser excitation, is many orders of magnitude greater than that provided by solar radiation at the Earth's surface (the maximum solar photon flux is about $10^{18} \text{ cm}^{-2} \text{ s}^{-1}$; assuming a carrier lifetime of 1 ns and an absorption coefficient of $1 \times 10^5 \text{ cm}^{-1}$, this translates into a photoinduced electron density of about 10^{14} cm^{-3} at steady state). Hence it is not possible to obtain slowed hot-carrier cooling in semiconductor QWs and superlattices with unconcentrated solar irradiation via a *hot* phonon bottleneck effect; solar concentration ratios greater than 10^4 would be required, resulting in severe practical problems.

However, the situation with three-dimensional confinement in quantum dots is more favourable. In the QD case, slowed hot-electron cooling is theoretically possible, even at arbitrarily low light intensity; this effect is simply called a phonon bottleneck, without the qualification of requiring hot phonons (*i.e.* a non-equilibrium distribution of phonons). More rigorous theoretical models for slowed cooling in quantised nanoparticles have been proposed by Bockelmann and coworkers (Bockelmann and Bastard, 1990; Bockelmann and Egeler, 1992) and Benisty and coworkers (Benisty, 1991; Benisty *et al.*, 1995). Thus, the slowed cooling could make the rate of other useful paths for the relaxation of highly energetic electron-hole pairs (*i.e.* excitons) competitive with cooling. One such important process is the creation of additional excitons in QDs when the absorbed photon energy is greater than twice the QD bandgap (HOMO-LUMO or $1S_h-1S_e$ transition). This process is known in bulk semiconductors as impact ionisation and is an inverse process to an Auger process, whereby one of two electron-hole pairs recombines non-radiatively and transfers this energy to the remaining electron or hole to create an excited electron-hole pair. In single QDs, PL blinking (intermittent PL as a function of time) has been observed and

has been explained (Nirmal *et al.*, 1996; Efros and Rosen, 1997) by this type of Auger process ionising one carrier over the potential barrier at the surface into the surface region. This creates an electric dipole in the QD that quenches subsequent radiative emission after subsequent photon absorption; after some time, the ionised carrier can return to the QD core and the PL is turned on again. Since this Auger process can occur in QDs, the inverse Auger process, whereby one high-energy electron-hole pair (created from a photon with $h\nu > E_g$) can generate two electron-hole pairs, was predicted by Nozik to also occur in QDs (Nozik, 1997, 2001b and 2002).

The proposed Benisty mechanism (Benisty *et al.*, 1991; Benisty, 1995) for slowed hot-carrier cooling and phonon bottleneck in QDs requires that cooling only occur via LO phonon emission. However, there are several other mechanisms by which hot electrons can cool in QDs. Most prominent among these is an Auger mechanism (Efros *et al.*, 1995) where the excess energy of the electron is transferred again to the hole, which then cools rapidly because of its larger effective mass and smaller energy level spacing. Thus, an Auger mechanism for hot-electron cooling can break the phonon bottleneck (Efros *et al.*, 1995). Other possible mechanisms for breaking the phonon bottleneck include electron-hole scattering (Vurgaftman and Singh, 1994), deep-level trapping (Sercel, 1995), and acoustical-optical phonon interactions (Inoshita and Sakaki, 1992 and 1997).

As will be shown below, some degree of slowed cooling (a partial phonon bottleneck) can occur in certain QDs and QD configurations, leading to cooling times of several ps compared with several hundred fs in bulk and certain other QDs (a factor of about 10). But we shall also see that the process of creating multiple excitons from single photons in QDs is so unexpectedly fast that it easily beats the cooling time without requiring the very long cooling times (ns to μ s) that would be expected from a full phonon bottleneck.

Experimental determination of cooling dynamics of phonon bottlenecks in QDs

Over the past several years, many investigations have been published that explore hot electron cooling/relaxation dynamics in QDs and the issue of a phonon bottleneck in QDs. The results are controversial, and there are many reports that both support and contradict the prediction of slowed hot-electron cooling in QDs and the existence of a phonon bottleneck: Nozik (2001b) gives a detailed reference list. One element of confusion is that while some of these publications report relatively long hot electron relaxation times (tens of ps) compared with what is observed in bulk semiconductors, the results are reported as being not indicative of a phonon bottleneck because the relaxation times are not excessively long and PL is observed (Heitz *et al.*, 1998; Li

and Arakawa, 1998; Sosnowski *et al.*, 1998). Theory predicts very long relaxation lifetimes (hundreds of ns to μ s) of excited carriers for the extreme, limiting condition of a full phonon bottleneck; thus the carrier lifetime would be determined by non-radiative processes and PL would be absent. However, since the interest here is on the rate of relaxation/cooling compared with the rate of electron separation and transfer, and multiple-exciton generation (MEG), we consider that slowed relaxation/cooling of carriers has occurred in QDs if the relaxation/cooling times are $>3\text{--}5$ ps (about an order of magnitude greater than that for bulk semiconductors). This is because electron separation and transport and MEG can be very fast (sub-ps). For solar fuel production, previous work that measured the time of electron transfer from bulk III–V semiconductors to redox molecules (metallocenium cations) adsorbed on the surface found that electron transfer (ET) times can also be sub-ps to several ps (Miller *et al.*, 1995; Meier *et al.*, 1997 and 1999; Diol *et al.*, 1998); hence photoinduced hot-carrier separation, transport and transfer can be competitive with electron cooling and relaxation if the latter is longer than about 10 ps. MEG rates can also be in the sub-ps regime (Ellingson *et al.*, 2005; Schaller *et al.*, 2005a).

In a series of papers, Sugawara *et al.* (Mukai *et al.*, 1996; Sugawara *et al.*, 1997; Mukai and Sugawara, 1998) have reported slow hot electron-cooling in self-assembled InGaAs QDs produced by Stranski–Krastanov (SK) growth on lattice-mismatched GaAs substrates. Using time-resolved PL measurements, the excitation-power dependence of PL, and the current dependence of electroluminescence spectra, these researchers report cooling times ranging from 10 ps to 1 ns. The relaxation time increased with electron energy up to the fifth electronic state. Also, Mukai and Sugawara (1999) have published an extensive review of phonon bottleneck effects in QDs, which concludes that the phonon bottleneck effect is indeed present in QDs.

In addition to studies on self-assembled SK type of QDs, studies of carrier cooling and relaxation have also been performed on colloidal QDs by Klimov *et al.* (Klimov and McBranch, 1998; Klimov *et al.*, 2000), Guyot-Sionnest *et al.* (1999), Ellingson *et al.* (2003), and Blackburn *et al.* (2003). These results (Guyot-Sionnest *et al.*, 1999) support the Auger mechanism for electron relaxation, whereby the excess electron energy is rapidly transferred to the hole, which then relaxes rapidly through its dense spectrum of states. When the hole is rapidly removed and trapped at the QD surface, the Auger mechanism for hot-electron relaxation is inhibited and the relaxation time increases. Klimov (2000) provides a review of this work.

Blackburn *et al.* (2003) and Ellingson *et al.* (2003) have reported results for the electron-cooling dynamics in InP QDs, where the QD surface was modified to affect hole trapping and also where only electrons were injected into the QD from an external redox molecule (sodium biphenyl) so that the holes necessary for the Auger

cooling mechanism were not present in the QD (Blackburn *et al.*, 2003). For InP, HF etching was found to passivate electronic surface states, but not hole surface states (Langof *et al.*, 2002; Mićić *et al.*, 2002); thus, holes can become localised at the surface in both etched and unetched TOPO-capped QDs, and the dynamics associated with these two samples will not deviate significantly. The relaxation was found to be bi-exponential and suggests the presence of two subsets of quantum dots within the sample (Blackburn *et al.*, 2003; Ellingson *et al.*, 2003). Since etching has been shown to passivate hole traps inefficiently, it was proposed that the two subsets of quantum dots should be probed in the experiment: one subset in which the hole and electron are efficiently confined to the interior of the nanocrystal (hole trap absent; exciton confined to the QD core), and one subset in which the hole is localised at the surface of the QD on a phosphorus dangling bond (hole trap present; charge-separated QD) (Blackburn *et al.*, 2003; Ellingson *et al.*, 2003).

With the electron and hole confined to the QD core, strong electron–hole interaction leads to efficient, fast relaxation via the Auger mechanism, and in QDs where the hole is localised at the surface, the increased spatial separation inhibits the Auger process and results in slower relaxation. The data imply that hole trapping at the intrinsic surface state occurs in less than 75 fs (Blackburn *et al.*, 2003).

To further investigate the mechanisms involved in the intraband relaxation, experiments were conducted in which only electrons were present in the QDs and holes were absent. Sodium biphenyl is a very strong reducing agent, which has been shown to inject electrons into the conduction band of CdSe QDs (Shim and Guyot-Sionnest, 2000; Shim *et al.*, 2001), effectively bleaching the 1S transition and allowing an IR-induced transition to the 1P_e level. Sodium biphenyl was therefore used to inject electrons into the 1S electron level in InP QDs (Blackburn *et al.*, 2003). This 1S_e electron may be excited to the 1P_e level with an IR pump and its relaxation dynamics monitored back to the ground 1S state. Time-resolved, IR-induced transitions in *n*-type (electron-injected) InP QDs show that the relaxation of the excited electrons from the 1P to the 1S level can be fitted to a single exponential, with an average time constant of 3.0 ps, corresponding to a relaxation rate of 0.092 eV ps⁻¹; in neutral 50 Å TOP/TOPO-capped InP QDs, the relaxation shows a large 400 fs component indicative of fast electron cooling. Similar conclusions were reported for electrons injected into ZnO and CdSe colloidal quantum dots (Guyot-Sionnest *et al.*, 2005). These experiments confirm that in the absence of a core-confined hole, electronic relaxation is slowed by about an order of magnitude. However, it should be noted that the relaxation rate in the absence of a hole is close to the relaxation rate with the hole localised at the surface. This is surprising and raises the question of why electron cooling in the absence of a hole is not longer. Possible explanations have

been proposed (Efros and Nozik, 2003) including that (1) positive counter ions of the oxidised sodium biphenyl are adsorbed on the QD surface and behave like a trapped hole in producing a significant Coulomb interaction with the electron to permit Auger cooling; and (2) an enhanced Huang-Rhys parameter occurs in charged QDs and enhances multiphonon relaxation.

Recent results obtained by Guyot-Sionnest *et al.* (2005) show that the nature of the surface ligands has a major effect on the relaxation dynamics. Depending on the surface ligand stabilising the QDs, the relaxation or cooling dynamics of hot excitons could be varied from 3.8 ps for tetradecylphosphonic acid, to 8 ps for oleic acid, 10 ps for octadecylamine, and 27 ps for dodecanethiol ligands. The later cooling rate is nearly two orders of magnitude slower than that for naked QDs or QDs capped with TOP-TOPO.

In contradiction to the results showing slowed cooling in QDs, many other investigations are reported in the literature in which a phonon bottleneck was apparently not observed. These results have been reported for both self-organised SK QDs and II–VI colloidal QDs: Nozik (2001b) gives a detailed reference list. However, in several cases (Heitz *et al.*, 1997; Heitz *et al.*, 1998; Sosnowski *et al.*, 1998), hot-electron relaxation was found to be slowed, but not sufficiently for the authors to conclude that this was evidence of a phonon bottleneck.

3.5 High conversion efficiency via multiple exciton generation in quantum dots

For a single-bandgap system, there are two fundamental ways in which hot carriers or excitons can be converted to useful work to enhance the efficiency of solar photon conversion. One way produces an enhanced photovoltage, and the other an enhanced photocurrent. As discussed above in Sections 3.1 and 3.4, the former requires that the hot carriers be extracted from the photoconverter before they cool (Boudreaux *et al.*, 1980; Ross and Nozik, 1982; Nozik, 2001b), while the latter requires the hot carriers to produce two or more electron–hole pairs (Kolodinski *et al.*, 1993; Landsberg *et al.*, 1993). This latter process is well known in bulk semiconductors and is termed *impact ionisation* (II); it is the inverse of the Auger process whereby two electron–hole pairs recombine to produce a single highly energetic electron–hole pair.

In order to achieve higher photovoltages, the rates of photogenerated carrier separation, transport and interfacial transfer across the semiconductor interface must all be fast compared with the rate of carrier cooling (Boudreaux *et al.*, 1980; Nozik, 1980; Williams and Nozik, 1984). The achievement of higher photocurrent requires that the rate of II (r_{II}) or electron–hole pair multiplication is greater than the rates of

carrier cooling (r_{cooling}), electron transfer ($r_{\text{ET}}^{\text{hot}}$) of hot electrons, and the forward Auger processes (r_{Auger}) (see Fig. 3.13). Also, the rate of electron transfer of cooled electrons (r_{ET}) must be faster than the radiative recombination rate (r_{rad}) and r_{Auger} . Thus the required inequalities are

$$r_{\text{MEG}} > r_{\text{ET}}^{\text{hot}}, r_{\text{cooling}}; r_{\text{Auger}} < r_{\text{MEG}}; r_{\text{ET}} > r_{\text{rad}} > r_{\text{Auger}}$$

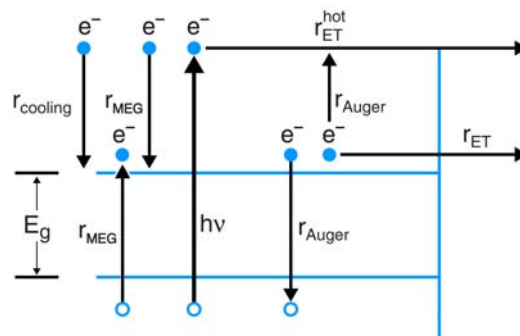


Figure 3.13 Various competing channels for hot-electron cooling. Source: Nozik (2005).

3.5.1 Cooling dynamics in quantum dots

As discussed above in Section 3.4, although the hot-carrier cooling rates are very fast in bulk semiconductors, the rates can be slowed down in quantised semiconductor structures because of the formation of discrete quantised energy levels. In quantum dots, photogenerated electrons and holes become bound to each other because of strong quantum confinement, and the correlated electron–hole pairs become excitons. The slowed cooling of energetic excitons can enhance the photon conversion efficiency by allowing electrical or chemical free energy to be extracted from the energetic excitons before they relax to their lowest electronic state and produce heat.

3.5.2 Electron–hole pair (exciton) multiplication in quantum dots

The formation of multiple electron–hole pairs per absorbed photon in photoexcited bulk semiconductors is a process typically explained by impact ionisation. In this process, an electron or hole with kinetic energy greater than the semiconductor bandgap produces one or more additional electron–hole pairs. The kinetic energy can

be created either by applying an electric field or by absorbing a photon with energy at least twice that of the semiconductor bandgap energy. The former is well studied and understood (Bude and Hess, 1992; Jung *et al.*, 1996; Harrison *et al.*, 1999). The latter process is less well studied, but has been observed in photoexcited *p-n* junctions of Si, Ge, and InSb (Tauc, 1959; Vavilov, 1959; Hodgkinson, 1963; Christensen, 1976).

However, impact ionisation has not contributed meaningfully to improved quantum yields in working solar cells, primarily because the II efficiency does not reach significant values until photon energies reach the ultraviolet region of the spectrum. In bulk semiconductors, the threshold photon energy for II exceeds that required for energy conservation alone because, in addition to conserving energy, crystal momentum must be conserved. Additionally, the rate of II must compete with the rate of energy relaxation by phonon emission through electron–phonon scattering. It has been shown that the rate of II becomes competitive with phonon scattering rates only when the kinetic energy of the electron is many times the bandgap energy (E_g) (Bude and Hess, 1992; Jung *et al.*, 1996; Harrison *et al.*, 1999). The observed transition between inefficient and efficient II occurs slowly; for example, in Si the II efficiency was found to be only 5% (*i.e.* total quantum yield = 105%) at $h\nu \approx 4$ eV ($3.6E_g$), and 25% at $h\nu \approx 4.8$ eV ($4.4E_g$) (Christensen, 1976; Wolf *et al.*, 1998). This large blue shift of the threshold photon energy for II in semiconductors prevents materials such as bulk Si and GaAs from yielding improved solar conversion efficiencies (Kolodinski *et al.*, 1993; Wolf *et al.*, 1998).

However, in quantum dots the rate of electron relaxation through electron–phonon interactions can be significantly reduced because of the discrete character of the electron–hole spectra, and the rate of Auger processes, including the inverse Auger process of exciton multiplication, is greatly enhanced owing to carrier confinement and the concomitantly increased electron–hole Coulomb interaction (see Fig. 3.14). Furthermore, crystal momentum need not be conserved because momentum is not a good quantum number for three-dimensionally confined carriers (from the Heisenberg Uncertainty Principle the well-defined location of the electrons and holes in the nanocrystal makes the momentum uncertain). Indeed, very efficient multiple electron–hole pair (multiexciton) creation by one photon has now been reported in six semiconductor QD materials: PbSe, PbS, PbTe, CdSe, InAs and Si. The first report was by Schaller and Klimov (2004) for PbSe NCs; they reported an excitation energy threshold for the efficient formation of two excitons per photon at $3E_g$, where E_g is the absorption energy gap of the nanocrystal (HOMO–LUMO transition energy). Schaller and Klimov reported a QY value of 218% (118% II efficiency) at $3.8E_g$; QYs above 200% indicated the formation of more than two excitons per absorbed photon. The NREL research group (Ellingson *et al.*, 2005) recently reported a QY value of 300%

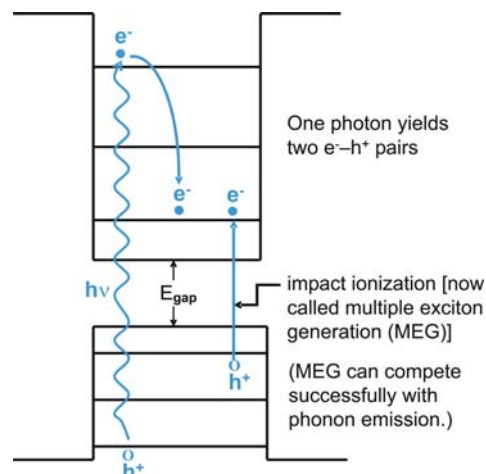


Figure 3.14 Multiple electron-hole pair (exciton) generation (MEG) in quantum dots.
Source: Nozik (2002).

for 3.9-nm diameter PbSe QDs at a photon energy of $4E_g$, indicating the formation of three excitons per photon for every photoexcited QD in the sample. The data also indicated that the sharp rise of MEG occurred more steeply and at a lower threshold photon energy with smaller-sized QDs. Evidence was also provided that the threshold energy for MEG by optical excitation is $2E_g$ (Ellingson *et al.*, 2005), and that efficient MEG also occurs in PbS (Ellingson *et al.*, 2005) and in PbTe nanocrystals (Murphy *et al.*, 2006). A new mechanism for MEG was introduced that invokes a coherent superposition of multiple excitonic states, meaning that multiple excitons are essentially created instantly on absorption of high-energy photons (Ellingson *et al.*, 2005; Shabaev *et al.*, 2006). This new model also explains the lower threshold photon energy for MEG at $2E_g$ for the IV-VI semiconductors (PbS, PbSe, PbTe) that have approximately equal effective masses for electrons and holes. This property led some authors (Schaller, 2004) to believe that the excess energy of the absorbed photon is split equally between the electron and hole, thus requiring a threshold of $3E_g$. However, several other models for MEG have also been proposed (Schaller *et al.*, 2005a; Franceschetti *et al.*, 2006).

Multiexcitons are detected by several spectroscopic measurements. The first one was to monitor the signature of multiexciton decay dynamics using transient absorption (TA) spectroscopy (Schaller and Klimov, 2004; Ellingson *et al.*, 2005; Schaller *et al.*, 2005b and 2006; Murphy *et al.*, 2006). The magnitude of the photo-induced absorption change at the band edge is proportional to the number of electron-

hole pairs created in the sample. In Ellingson *et al.* (2005) and Murphy *et al.* (2006), the transients are detected by probing either with a band edge (energy gap or HOMO–LUMO transition energy $\equiv E_g$) probe pulse, or with a mid-IR probe pulse that monitors intraband transitions in the newly created excitons. Our multiple exciton Auger recombination analysis relies only on data for delays >5 ps, by which time carrier multiplication and cooling are complete.

In Ellingson *et al.* (2005) and Murphy *et al.* (2006), the dependence of the MEG QY on the ratio of the pump photon energy to the bandgap (E_{hv}/E_g) varied from 1.9 to 5.0 for PbSe QD samples with $E_g = 0.72$ eV (diameter = 5.7 nm), $E_g = 0.82$ eV (diameter = 4.7 nm), and $E_g = 0.91$ eV (diameter = 3.9 nm), as shown in Fig. 3.15. For all three samples, the sharp rise in QY begins at about three times the energy gap. For the two PbSe samples with $E_g = 0.82$ eV (4.7 nm diameter) and 0.72 eV (5.7 nm diameter), it is estimated that a QY of 300% is reached at an E_{hv}/E_g value of 5.5. It was noted that the $2P_h$ – $2P_e$ transition in the QDs is resonant with the $3E_g$ excitation, corresponding to the sharp onset of increased MEG efficiency. If this symmetric transition ($2P_h$ – $2P_e$) dominates the absorption at $\sim 3E_g$, the resulting excited state provides both the electron and the hole with excess energy of $1E_g$, in resonance with the lowest exciton absorption (at $1E_g$). These data also showed that the QY begins to surpass 100% at E_{hv}/E_g values greater than 2.0 (see Fig. 3.15). We carefully measured sixteen QY values between $2.1E_g$ and $2.9E_g$ (mean value = 109.8%) and eleven QY values between $1.2E_g$ and $2.0E_g$ (mean value = 101.3%). Application of statistical t-

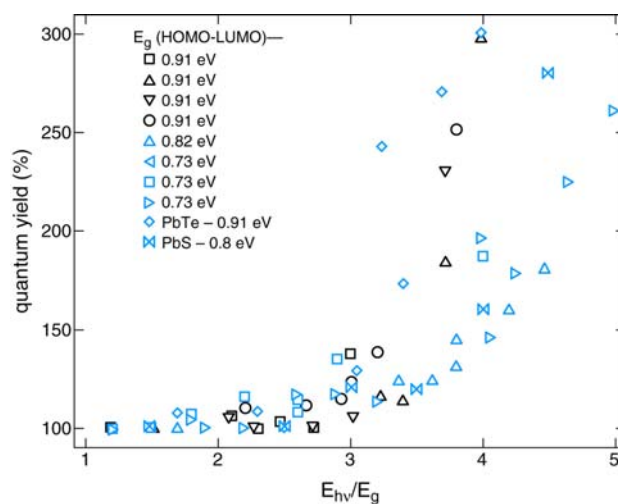


Figure 3.15 Quantum yield of MEG vs photon energy (E_{hv}/E_g) for PbSe, PbS, and PbTe QDs. The values for QD bandgaps (HOMO–LUMO) are listed next to the symbols.

tests showed that the QY values for photon energies between $1E_g$ and $2E_g$ were not statistically different from 100% (P value = 0.105), while the difference in QYs between $1.2E_g$ – $2.0E_g$ and $2.1E_g$ – $2.9E_g$ were very statistically significant with a P value of 0.001 (Ellingson *et al.*, 2005). Also, simple visual inspection of Fig. 3.15 shows a significant difference between the QY values between $1E_g$ – $2E_g$ and $2E_g$ – $3E_g$.

Additional experiments observing MEG have been reported for CdSe (Schaller *et al.*, 2005b), PbTe (Murphy *et al.*, 2006) and InAs (Pijpers *et al.*, 2007). In addition to transient absorption studies, these optical experiments use time-resolved photoluminescence, terahertz spectroscopy, and quasi-CW spectroscopy.

3.5.3 Theory of multiple exciton generation

In a novel recent theory of MEG (Shabaev *et al.*, 2006), efficient MEG by a single photon in NCs is explained by the formation of a coherent superposition of single- and multi-exciton states (see Fig. 3.16). The model uses a time-dependent density matrix approach, which allows simultaneous consideration of an arbitrary strength of coupling between single- and multi-exciton states, different dephasing rates for these states, and short pulse excitation of the NCs. The steady-state solution of the density matrix equations yields the following conditions for efficient MEG: (1) the energy relaxation rate of a single exciton (γ_1) initiated by light must be slower than both the energy relaxation rate of the multiexciton (γ_2) and the rate of Coulomb coupling between the two states ($W_C/\hbar$), where W_C is the matrix element of the Coulomb interaction between the single- and multi-exciton states:

$$W_C = \langle \psi_{1\text{Ex}} | q^2 / \epsilon r | \psi_{n\text{Ex}} \rangle \quad (3.25)$$

and $\psi_{1\text{Ex}}$ is the initial single-exciton state, $\psi_{n\text{Ex}}$ is the multiexciton state, ϵ is the permittivity of the QD, and $\mathbf{r}$ is the vectorial distance between the electronic particles in the QD.

In the case of strong coupling, $\gamma_2 < W_C/\hbar$, MEG by a short light pulse is accompanied by quantum beats with the frequency $W_C/\hbar$. These oscillations are damped in the opposite case of weak coupling, $\gamma_2 > W_C/\hbar$; however, the MEG process can still be very efficient. The density matrix approach is the only self-consistent method that takes into account the diverse processes responsible for MEG in NCs.

Under optical excitation of NCs, a single photon can only generate a *single* electron–hole pair which is not an eigenstate of the multielectron Hamiltonian that describes the superposition of multiexcitons. This initial single electron–hole pair

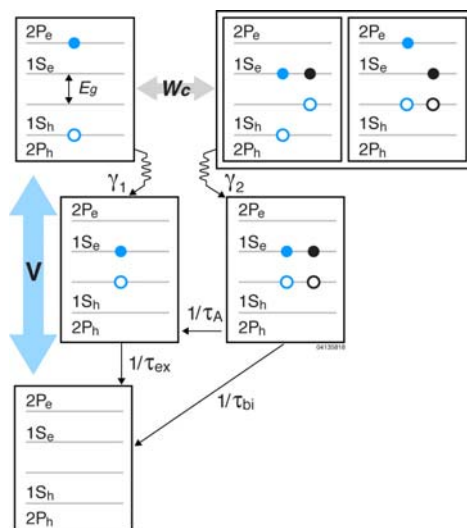


Figure 3.16 Theoretical model for MEG. For simplicity, only three degenerate states are shown following absorption of a $3E_g$ photon ($2P_h - 2P_e$): the initial single $2P_h2P_e$ exciton, and the two excited biexciton states ($2P_h1S_e1S_h1S_e$, $1S_h1S_e2P_e1S_e$). For simplicity, the fourth state ($1S_h1S_e1S_h1S_e$) is not represented and electronic states between the 1S and 2P states are also not shown. The three degenerate states are coupled by the Coulomb interaction (W_c) and form a coherent superposition of three states; their populations experience oscillations with time (quantum beats). The relaxation (cooling) rate of the single exciton state ($1/\gamma_1$) is slower than the relaxation rate of the biexciton states ($1/\gamma_2$) because of the asymmetric charge distribution in the excited biexcitons (the latter enhances electron-phonon coupling). This preferential dephasing of the coherence for the excited biexcitons results in the final formation of a biexciton from a single photon. V is the overall rate of decay of the excited single exciton to the ground state, while $1/\tau_{ex}$ and $1/\tau_{bi}$ are the rates of the decay of the relaxed (cooled) single exciton and biexcitons to the ground state. All of these latter three rates are slow compared with the rates of relaxation and Coulomb coupling. Source: Shabaev *et al.* (2006).

state (exciton) is coupled with the multiexciton states and creates a superposition of single and multiexcitons. Below, the theory is applied to PbSe NCs, in which the onset of very efficient MEG occurs at photon energies of $3E_g$, corresponding to a $2P_h \rightarrow 2P_e$ transition (Shabaev *et al.*, 2006) (but note that the threshold for the onset of MEG is $2E_g$).

For a strongly coupled superposition of single- and multi-exciton states, the question of MEG efficiency is reduced to the question of the state relaxation. The single- and multi-exciton components of the superposition have different decay rates, and the ratio of these rates determines what a single photon creates: a single exciton or multiexcitons.

To calculate the efficiency of MEG, it is necessary to compare the populations of the ground state (band-edge) biexcitons, N_{bi} , to the single exciton, N_{ex} , during time, t , which is much longer than the relaxation times of the excited states ($1/\gamma_1$ and $1/\gamma_2$, respectively) but much shorter than the lifetime of the band-edge excitons ($1/T_{ex}$ and $1/T_{bi}$); *i.e.* $1/\gamma_1 \leq 1/\gamma_2 \ll 1/T_{bi} \ll 1/T_{ex}$. This approach ultimately yields the ratio of the band-edge biexciton and single-exciton populations (Shabaev *et al.*, 2006)

$$N_{bi} / N_{ex} = (\gamma_2 / \gamma_1) P_{1-2} \quad (3.26)$$

where P_{1-2} is the probability of the population of the excited biexciton state via its Coulomb coupling to the excited single-exciton state created by a photon; the value of P_{1-2} is always < 1 .

Equation 3.26 shows that efficient MEG, *i.e.* the predominant generation of multiexcitons, is possible only if the relaxation rate of the biexciton is much faster than that of the exciton ($\gamma_2 \gg \gamma_1$), because $P_{1-2} < 1$. For strong coupling, the transition probability approaches unity and the population ratio approaches its maximum value of (γ_2/γ_1) . This result has a transparent physical meaning: for a strongly coupled superposition state, the populations of N_{bi} and N_{ex} are controlled by the state decay into the two independent thermalisation channels.

MEG, however, can be efficient even for a weak coupling regime, $W_C/\hbar \ll \gamma_2$. In this case $N_{bi}/N_{ex} = 4W_C^2/(\hbar^2 \gamma_1 \gamma_2)$, and can be much larger than unity if $W_C/\hbar \gg \gamma_1$ (Shabaev *et al.*, 2006). The relative population N_{bi}/N_{ex} is then determined by the ratio of the biexciton creation rate, $(2W_C/\hbar)^2\gamma_2$, to the direct single-exciton relaxation rate, γ_1 .

As discussed above, MEG has been observed by measuring the bleaching of the optical absorption at the bandgap frequency. Although the initially excited exciton, whose energy is above the MEG threshold, does not directly populate the states at the bandgap, the bleaching increases rapidly after the excitation. The timescale of the bleaching evolution provides information about the strength of the coupling between the initially excited state and the multiexcitons which populate the low-energy states at the bandgap. Efficient MEG leads to bleaching that significantly exceeds the bleaching of the single band-edge exciton. In the excitation frequency range for the $2P_h \rightarrow 2P_e$ transition, the time dependence of the relative change (bleach) of the band-edge absorption after a short pulse excitation is shown in Fig. 3.17. This figure shows that strong coupling ($W_C/\hbar > \gamma_2$) significantly shortens the bleach rise time ($\sim \hbar/W_C$) and leads to quite pronounced oscillations (quantum beats) resulting from the creation of the coherent superposition of the single- and bi-exciton excited states. The coherent oscillations become overdamped even for a relatively strong coupling if $W_C/\hbar \sim \gamma_2$,

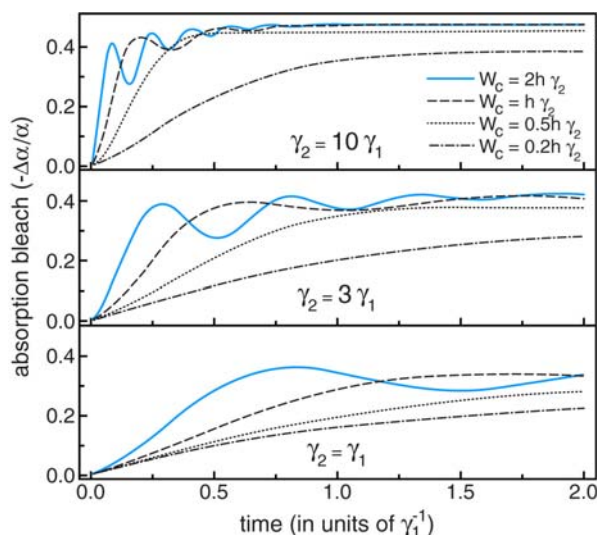


Figure 3.17 Time evolution of the $1S_e 1S_h$ exciton population as a function of the relaxation times and Coulomb coupling for the single and biexcitons. The rise time of the biexciton (formation time) and the presence of strong or damped quantum beating depends upon the relative values of the W_c , γ_1 , and γ_2 as shown above. Source: Shabaev *et al.* (2006).

and the coupling still substantially mixes the exciton states. In the weak coupling regime, $W_c/\hbar < \gamma_2$, the bleach rise time is slowed down by the biexciton decay rate and is proportional to $\hbar^2 \gamma_2 / W_c^2$.

Detuning of the resonance between single- and multi-exciton states, for example from a significant size dispersion of the NC population, also affects the bleach rise-time behaviour. Generally, the detuning increases the frequency of the beats and slows down the formation of a coherent superposition. Thus, a significant NC size dispersion will wash out pronounced quantum beats and slow down the average rise time of the bleach signal. This appears to be the case for the rise-time data in Ellingson *et al.* (2005), where only an apparent half-oscillation is observed.

Although the major mechanism for carrier thermalisation in NCs is still not agreed upon, it is found theoretically (Shabaev *et al.*, 2006) that the biexciton relaxation rate is much faster than the single-exciton relaxation rate, assuming that the relaxation rate for various electron-hole pair configurations is proportional to their coupling with phonons. These calculations show that polar interactions of intrinsic semiconductor phonons in CdSe and PbSe NCs with asymmetric electron-hole pair configurations are 10–40 times stronger than that of their coupling with symmetric electron-hole pair

configurations ($nL_e nL_h$). This is because in the latter case, charge distributions of the optically created electron and hole compensate each other almost exactly at each point of the NC, and the NC thus retains its local neutrality even after exciton creation. As a result, exciton interactions with the polar optical phonons that are sensing the total charge are very weak (Schmitt-Rink *et al.*, 1987). It is also quite obvious that the coupling of asymmetric electron–hole configurations with phonons of organic molecules at the NC surface (Guyot-Sionnest *et al.*, 2005) should be significantly stronger as well. In both cases, weak coupling of symmetric electron–hole pairs created by light with phonons suppresses their relaxation and would result in efficient MEG because the condition $\gamma_1 \ll \gamma_2$ is fulfilled.

Efficient MEG also requires strong coupling between exciton and biexciton states relative to γ_1 : $W_C/\hbar \gg \gamma_1$. Fulfilling this condition satisfies the obvious requirement that the exciton thermalisation time should be longer than the coupling time between the optically created exciton and the asymmetric biexciton state (Nozik, 2002).

Experimental results indicate that $1/\gamma_1 \sim 2\text{--}6$ ps, depending on the NC size (Guyot-Sionnest *et al.*, 1999; Ellingson *et al.*, 2005; Harbold *et al.*, 2005; Schaller *et al.*, 2005c), and Shabaev *et al.* (2006) reported that the condition $W_C/\hbar \gg \gamma_1$ is fulfilled even for the lowest estimation of W_C . In NCs, the dielectric constant, $\epsilon(a)$, is smaller than in bulk, and consequently the coupling W_C is stronger.

Theoretically, the MEG QY will not show a size dependence if it is assumed that the ratio γ_1/γ_2 does not depend on size. At the same time, an increase of γ_2/γ_1 always leads to an increase of QY. The predicted size independence of the average number of electron–hole pairs generated via excitation of the same optical transition is consistent with experimental results, which show that the QY depends only weakly on size and is a function primarily of only one parameter: E_{hv}/E_g (Schaller and Klimov, 2004; Ellingson *et al.*, 2005; Murphy *et al.*, 2006; Schaller *et al.*, 2006).

3.5.4 Thermodynamic calculations of conversion efficiency in MEG QD solar cells

Detailed balance model

Hanna and Nozik (2006) have used the detailed balance model to calculate the power conversion efficiency of single-gap and two-gap tandem solar conversion devices which employ QD absorbers capable of MEG after photon absorption. The detailed balance model has been used previously (Shockley and Queisser, 1961; Werner *et al.*, 1994; Spirkel and Ries, 1995; Brendel *et al.*, 1996; Würfel, 1997; de Vos and Desoete, 1998; Landsberg and Badescu, 2002) to calculate the limiting efficiency of ideal solar

cells with a single bandgap having greater than unity QY resulting from the process of impact ionisation. We will apply this model to calculate the efficiency of single-gap and two-gap tandem PV and photoelectrochemical (PEC) devices having various combinations of MEG absorbers. Here, the designation PEC refers to a solar-driven photoelectrolysis cell for the production of H₂ from H₂O.

The current-versus-voltage dependence for a single-threshold device is written as

$$I(V, E_g) = I_G(E_g) - I_R(V, E_g) \quad (3.27)$$

where I_G is the photogenerated current, I_R is the recombination current associated with radiative emission, E_g is the absorption threshold or bandgap of the absorber, and V is the photovoltage generated by the cell. Expressions for the photogenerated current I_G and recombination current I_R for a single-gap cell are written as

$$I_G(E_g) = q \int_{E_g}^{E_{\max}} QY(E) \Gamma(E) dE \quad (3.28)$$

$$I_R(V, E_g) = qg \int_{E_g}^{\infty} \frac{QY(E) E^2}{\exp\{[E - qQY(E)V]/kT\} - 1} dE \quad (3.29)$$

where E is the photon energy, q is the electronic charge, k is Boltzmann's constant, T is the temperature of the device (here $T = 300$ K), and $g = 2\pi/c^2 h^3$, where c is the speed of light in vacuum and h is Planck's constant. The quantum yield, $QY(E)$, allows for the generation and recombination of multiple charge pairs per photon over the appropriate energy range. The unconcentrated ASTM G-173-3 solar spectrum¹ is used as the illumination source; $\Gamma(E)$ is the photon flux associated with the AM1.5G spectrum. $E_{\max}$ is the maximum photon energy in the solar spectrum; for AM1.5G, $E_{\max} = 4.428$ eV. For practical purposes, $E_{\max} \sim 4$ eV, because the integrated solar current above 4 eV in the standard AM1.5G spectrum is only $\sim 5 \mu\text{A cm}^{-2}$. In eq. 3.28, we neglect carrier generation from ambient blackbody radiation, which is a good approximation for $E_g \geq 0.2$ eV. Implicit in eqs. 3.27–3.29 are the assumptions of the detailed balance model: all photons with energy above the absorption threshold are absorbed, the quasi-Fermi level separation is constant and equal to qV across the device (this is equivalent to the assumption of infinite carrier mobility), and the only active recombination mechanism is radiative recombination.

MEG is incorporated into the model through the energy-dependent quantum yield, $QY(E)$, for photon to carrier generation, which may exceed one over certain photon

¹ The ASTM G-173-3 Reference Solar Spectra can be found at <http://rredc.nrel.gov/solar/spectra/am1.5/>.

energy ranges. The functional form for $QY(E)$ to model the quantum yield for ideal MEG QD absorbers is given by a sum of step functions

$$QY(E) = \sum_{m=1}^M \theta(E, mE_g) \quad (3.30)$$

where $\theta(E, mE_g)$ is the Heaviside unit step function. Taking $M = 1$ gives the usual assumption of one electron–hole pair generated on absorption of one photon. Setting $M = M_{\max} = E_{\max}/E_g$ gives the maximum possible multiplication allowed by energetic constraints and the highest possible efficiency for a MEG device. In the following, we denote MEG absorbers with $M = 1$, $M = 2$, and $M = M_{\max}$ as M1, M2, and $M_{\max}$, respectively.

The photovoltaic power conversion efficiency $\eta_{\text{pv}}(V)$ at a given operating point (I, V) is given by

$$\eta_{\text{pv}}(V) = I(V)V / P_{\text{in}} \quad (3.31)$$

where $P_{\text{in}} = 100 \text{ mW/cm}^2$ is the integrated optical power in the AM1.5G spectrum. The photoelectrochemical conversion efficiency for the production of stored chemical energy as H_2 from water splitting is written as

$$\eta_{\text{H}_2}(V) = I(V)U_{\text{H}_2} / P_{\text{in}} \quad (3.32)$$

where $U_{\text{H}_2} = 1.23 \text{ V}$ is the minimum thermodynamic potential required for water splitting at 300 K. In actual water-splitting devices, the operating or bias point of the cell V will be larger than U_{H_2} by the sum of the anode and cathode overpotentials and the resistive potential drop of the electrolyte. We will use V_{over} to denote the sum of these overpotentials (losses) and assume that it is independent of current. Then the operating voltage is

$$V = V_{\text{over}} + U_{\text{H}_2} \quad (3.33)$$

The maximum efficiency for a single-gap device with a given absorption threshold and QY can be found from eqs. 3.27–3.31 by maximising the efficiencies in eqs. 3.32 and 3.33 with respect to the operating voltage V . For two-gap tandem devices, the top and bottom cells are connected in series optically, but they can be connected electrically either in series or parallel. The subscript 1 denotes parameters for the top cell and 2 for the bottom cell. For a series-connected tandem device, the total current must be the same in each cell, while the photovoltage across the device is the sum of the voltages developed across each cell. For a parallel-connected tandem device, the photovoltage is equal to the voltage across the sub-cells, and the total current is the sum of the currents generated by each cell. For both PV and water-splitting tandem

devices, a restriction on the possible combinations of E_1 and E_2 arises from the requirement that a portion of the solar spectrum must be absorbed in the bottom cell for a tandem device to function. For all absorber combinations of top and bottom cells, the absorption threshold for the top cell must be larger than the absorption threshold of the bottom cell. Additionally, a water-splitting tandem device must have a combination of threshold energies E_1 , E_2 that generates an open-circuit voltage greater than $V_o + U_{H_2}$ for the device to be able to split water.

Single-bandgap PV cells

The efficiency of ideal single-gap PV devices with carrier multiplication arising from an impact ionisation process has been reported by several authors (Werner *et al.*, 1994; Spirkel and Ries, 1995; Brendel *et al.*, 1996; Würfel, 1997; de Vos and Desoete, 1998; Landsberg and Badescu, 2002). In these publications, blackbody solar illumination was assumed for the solar spectrum. Under the AM1.5G spectrum (integrated incident solar power = 100 mW cm^{-2}), the maximum efficiency for single-gap cells operating at 300 K with the quantum yields of eqs. 3.31–3.33 is shown in Fig. 3.18. The curve labelled M_1 has no carrier multiplication and corresponds to the Shockley–Queisser limit with a maximum efficiency 33.7% occurring at $E_g = 1.34 \text{ eV}$.² The efficiency curve labelled M_{max} is calculated using the maximum possible integral number of multiplications consistent with the constraint imposed by the available excess energy $M_{\text{max}} = E_{\text{max}}/E_g$. The maximum efficiency for a single-gap

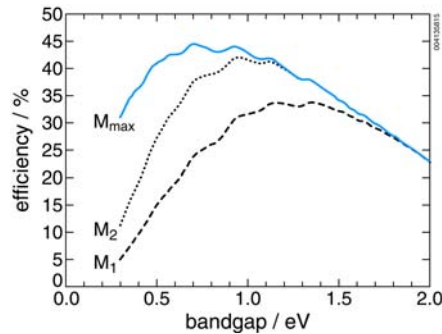


Figure 3.18 PV conversion efficiency as a function of QD bandgap for single-gap devices with the following absorber types: M_1 (no exciton multiplication), M_2 (two excitons per photon above $h\nu = 2E_g$), and M_{max} (maximum possible number of excitons per photon from solar spectrum).

² Values of the Shockley–Queisser limiting efficiency differing by a few absolute percent appear in the literature because of differences in the incident spectra or operating temperature used in the calculations.

MEG PV device is 44.4% for bandgap $E_g = 0.7$ eV, where $M_{\max} = 6$. This is similar to the value reported by de Vos and Desoete (1998) for a blackbody-illuminated PV device with maximum carrier multiplication. It is interesting to note that a high efficiency of $\sim 42\%$ is still obtained with a multiplication of only 2 (curve labelled M_2), which is 94% of the absolute maximum. This means that high multiplication values (>2) are not essential for substantially increasing the efficiencies of single-gap PV devices having carrier multiplication absorbers.

Tandem cells with two bandgaps for water splitting

Calculations by Hanna and Nozik (2006) show that the conversion efficiency of two-gap tandem cells for PV that exhibit MEG does not improve significantly compared with two-gap tandem cells with no MEG occurring in any of the tandem layers. The maximum value for all two-gap tandem combinations is $\sim 45\%$ at one Sun. For water-splitting cells with no MEG, the maximum efficiency (with zero overvoltage) increases from $\sim 31\%$ using a single-gap photoelectrode to $\sim 41\%$ using a two-gap tandem cell configuration. The optimum bandgaps for the latter case at zero overvoltage are 1.4 eV for the top cell and 0.5 eV for the bottom cell; for the former case, the optimum bandgap is 1.31 eV for zero overvoltage. For an overvoltage V_{over} of 0.6 V, the maximum efficiency for a single-gap cell is 13%, and for the tandem cell it is 33%. The optimum bandgaps for these two cases are 2.13 eV for the single-gap cell, and 1.6 eV and 0.9 eV for the tandem cell. However, the efficiencies of both single-gap and tandem photoelectrolysis cells increase only modestly with MEG.

Conclusions about conversion efficiencies with MEG solar cells

Solar conversion devices with MEG absorbers have in some cases higher limiting efficiencies than devices without MEG. The largest increase in efficiency over the case with no MEG occurs for single-gap PV devices, where the limiting efficiency increases by about one third, from 33.7% to 44.4% for maximum carrier multiplication. An increase in efficiency from 30.6% to 32.0% occurs when incorporating a M_2 MEG absorber into a single-gap photoelectrolysis cell. For the single-gap PV efficiency, increasing the multiplication above M_2 will not improve the efficiency of a single-gap photoelectrolysis device above that obtained with M_2 . The smaller improvement in efficiency of single-gap photoelectrolysis devices is due to the additional constraint imposed on the photovoltage by the minimum free energy required for water splitting. In PV devices, the voltage at the maximum power point is not restricted as for the photoelectrolysis devices, allowing higher efficiency to be

obtained at lower bandgaps where the increased current from MEG outweighs the reduction in voltage.

In tandem PV devices, the inclusion of MEG in either top or bottom cells improves the efficiency when the top cell has an M2 absorber, but the increase is relatively small—only about 2%. A MEG absorber improves the efficiency of a tandem photoelectrolysis device to a greater degree than a tandem PV device, although the efficiency of the photoelectrolysis device is slightly lower than the PV device with the same configuration. The introduction of overvoltage losses in photoelectrolysis devices with MEG shows that they lose their advantage over normal photoelectrolysis devices for overpotentials above 0.4 V. At low overpotentials, single-gap and tandem devices with MEG absorbers have higher efficiencies than devices without MEG. For single-gap devices, the efficiency of the M1 and M2 devices becomes equal above $V_{\text{over}} = 0.4$ V. As V_{over} increases, larger bandgaps for both the top and bottom cells in a tandem device are required to obtain sufficient photovoltage to overcome the losses and drive the water-splitting reaction. In general, careful analyses of the efficiencies are required before concluding that a solar converter with a MEG absorber will perform better than a device without MEG.

3.6 Quantum dot solar cell configurations

There are two fundamental approaches for enhancing the solar photon conversion efficiency: increased photovoltage (Boudreaux *et al.*, 1980; Ross and Nozik, 1982) and increased photocurrent (Kolodinski *et al.*, 1993; Landsberg *et al.*, 1993). These can be accessed, in principle, in at least three different QD solar cell configurations; these configurations are shown in Fig. 3.19 and described below.

3.6.1 Photoelectrodes composed of quantum dot arrays

In this configuration, the QDs are formed into an ordered three-dimensional array with inter-QD spacing sufficiently small such that strong electronic coupling occurs and minibands are formed to allow long-range electron transport (see Fig. 3.19a). The optimum shape of QDs for this configuration is in the form of cubes rather than spheres, in order to get all inter-QD surfaces as close as possible. This system is a 3-D analogue to a one-dimensional superlattice and the miniband structures described above in Section 3.2 (see Fig. 3.4).

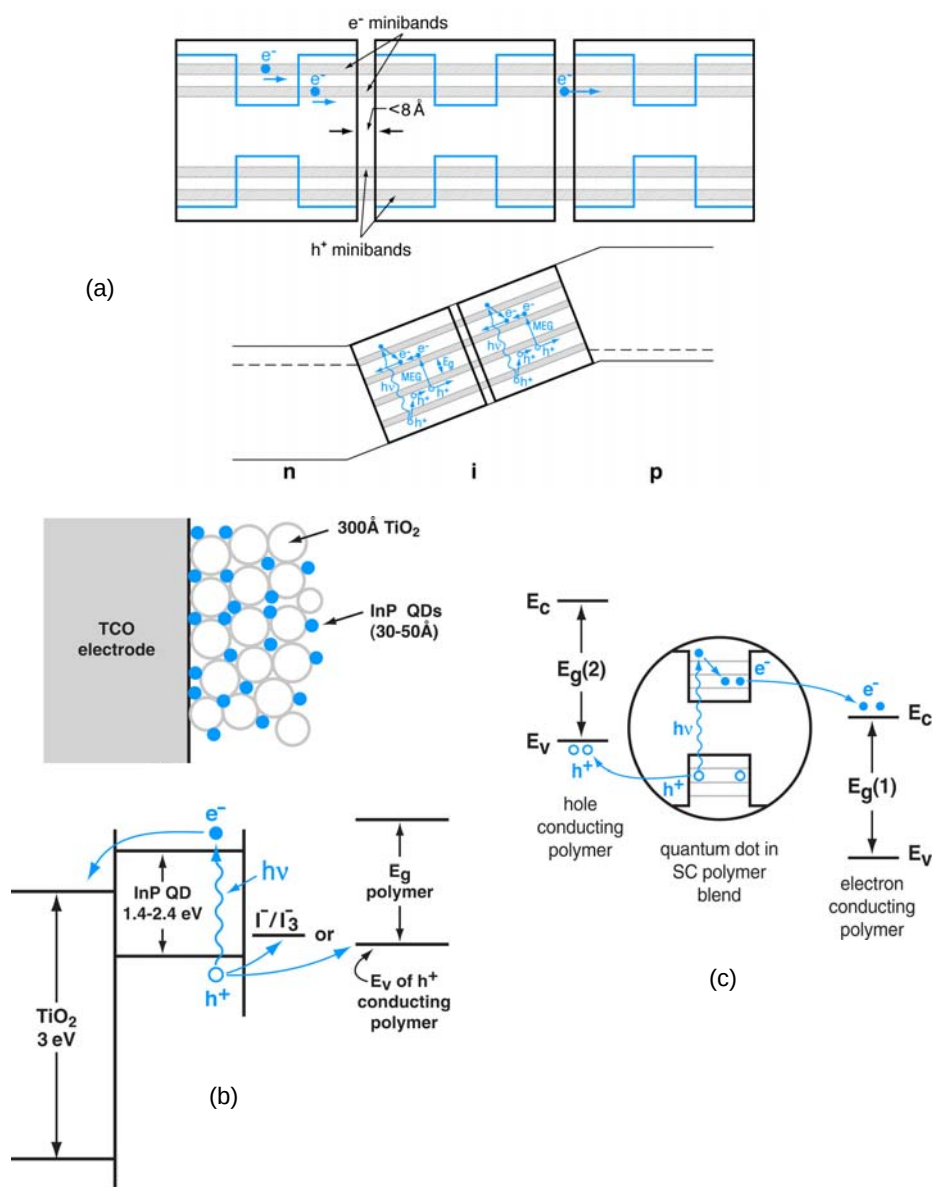


Figure 3.19 Configurations for QD solar cells. (a) A QD array used as the *i* region of a *p-i-n* junction cell configuration; (b) QDs used to sensitise a nanocrystalline film of TiO₂ to visible light. This configuration is analogous to the dye-sensitised solar cell with the dye replaced by QDs; (c) QDs dispersed in a blend of electron- and hole-conducting polymers. In these configurations the occurrence of MEG could produce higher photocurrents and higher conversion efficiency. Source: Nozik (2002).

The delocalised quantised 3-D miniband states could be expected to slow the carrier cooling and permit the transport and collection of hot carriers to produce a higher photopotential in a PV cell or in a photoelectrochemical cell where the 3-D QD array is the photoelectrode (Nozik, 1996). Also, MEG might be expected to occur in the QD arrays, enhancing the photocurrent (see Fig. 3.14). However, hot-electron transport/collection and MEG cannot occur simultaneously; they are mutually exclusive and only one of these processes can be present in a given system.

Significant progress has been made in forming cubic and spherical 3-D arrays of both colloidal (Mićić *et al.*, 1998 and 2001; Murray *et al.*, 2000) and epitaxial (Sugawara, 1999) II–VI and III–V QDs. The former have been formed via evaporation and crystallisation of colloidal QD solutions containing a uniform QD size distribution; crystallisation of QD solids from broader size distributions leads to close-packed QD solids, but with a high degree of disorder. Concerning the latter, arrays of epitaxial QDs have been formed by successive epitaxial deposition of epitaxial QD layers; after the first layer of epitaxial QDs is formed, successive layers tend to form with the QDs in each layer aligned on top of one other (Nakata *et al.*, 1999; Sugawara, 1999). Theoretical and experimental studies of the properties of QD arrays are currently under way. Major issues are the nature of the electronic states as a function of inter-dot distance, array order *versus* disorder, QD orientation and shape, surface states, surface structure/passivation, and surface chemistry. Transport properties of QD arrays are also of critical importance. Initial results show significant transport in close-packed QD arrays (Beard *et al.*, 2003; Murphy *et al.*, 2006).

3.6.2 *Quantum dot-sensitised nanocrystalline TiO₂ solar cells*

This configuration is a variation of the promising new type of photovoltaic cell discussed above and in Chapter 8 which is based on dye sensitisation of nanocrystalline TiO₂ layers (Moser *et al.*, 1998; Grätzel, 2000; Hagfeldt and Grätzel, 2000). In this cell, dye molecules are chemisorbed onto the surface of 10–30 nm-size TiO₂ particles that have been sintered into a highly porous nanocrystalline 10–20 μ m TiO₂ film. On photoexcitation of the dye molecules, electrons are very efficiently injected from the excited state of the dye into the conduction band of the TiO₂, effecting charge separation and producing a photovoltaic effect. The cell circuit is completed using a non-aqueous redox electrolyte that contains a I[−]/I₃[−] redox relay and a Pt counter electrode to allow reduction of the adsorbed photooxidised dye back to its initial non-oxidised state (via I₃[−] produced at the Pt cathode by reduction of I[−]).

For the QD-sensitised cell, QDs are substituted for the dye molecules, as shown in Fig. 3.19b; they can be adsorbed from a colloidal QD solution (Zaban *et al.*, 1998) or produced *in situ* (Weller, 1991; Liu and Kamat, 1993; Vogel *et al.*, 1994; Hoyer and Könenkamp, 1995). Successful PV effects in such cells have been reported for several semiconductor QDs including InP, CdSe, CdS, PbS and InAs (Weller, 1991; Liu and Kamat, 1993; Vogel *et al.*, 1994; Hoyer and Könenkamp, 1995; Zaban *et al.*, 1998; Nozik, 2001a; Robel *et al.*, 2006; Yu *et al.*, 2006; Tachibana *et al.*, 2007). Possible advantages of QDs over dye molecules are the tunability of optical properties with size, better heterojunction formation with solid hole conductors, and the unique potential capability of the QD-sensitised solar cell to produce quantum yields greater than one by MEG (inverse Auger effect) (Nozik, 2002).

3.6.3 *Quantum dots dispersed in organic semiconductor polymer matrices*

Photovoltaic effects have been reported in structures consisting of QDs forming junctions with organic semiconductor polymers. In one configuration, a disordered array of CdSe QDs is formed in a hole-conducting polymer—MEH-PPV (poly(2-methoxy, 5-(2'-ethyl)-hexyloxy-*p*-phenylenevinylene) (Greenham *et al.*, 1996). On photoexcitation of the QDs, the photogenerated holes are injected into the MEH-PPV polymer phase and are collected via an electrical contact to the polymer phase. The electrons remain in the CdSe QDs and are collected through diffusion and percolation in the nanocrystalline phase to an electrical contact to the QD network. Initial results showed relatively low conversion efficiencies (Greenham *et al.*, 1996 and 1997; Choi *et al.*, 2006; Cui *et al.*, 2006), but Huynh *et al.* (1999) reported improvements with rod-like CdSe QD shapes embedded in poly(3-hexylthiophene) (the rod-like shape enhances electron transport through the nanocrystalline QD phase). In another configuration (Arango *et al.*, 1999), a polycrystalline TiO₂ layer is used as the electron-conducting phase, and MEH-PPV is used to conduct the holes; the electron and holes are injected into their respective transport media on photoexcitation of the QDs.

A variation of these configurations is to disperse the QDs into a blend of electron- and hole-conducting polymers (see Fig. 3.19c). This scheme is the inverse of light-emitting diode structures based on QDs (Colvin *et al.*, 1994; Dabbousi *et al.*, 1995; Schlamp *et al.*, 1997; Mattoussi *et al.*, 1998; Mattoussi *et al.*, 1999). In the PV cell, each carrier-transporting polymer would have a selective electrical contact to remove the respective charge carriers. A critical factor for success is to prevent electron–hole recombination at the interfaces of the two polymer blends; prevention of electron–hole recombination is also critical for the other QD configurations mentioned above.

All of the possible QD-organic polymer photovoltaic cell configurations would benefit greatly if the QDs could produce multiple electron-hole pairs via MEG. This is also true for all the QD solar cell systems described in Section 3.6. The various cell configurations simply represent different modes of separating, collecting and transporting the photogenerated carriers produced in the QDs.

3.7 Summary and conclusions

Size quantisation in semiconductors greatly affects the relaxation dynamics of photo-induced carriers. These effects include slowed hot electron relaxation (phonon bottleneck) and enhanced Auger processes, including very efficient multiple exciton generation (MEG). The theoretical and measured energy threshold for carrier multiplication (impact ionisation) in bulk semiconductors (*e.g.* Si and Ge) is 4–5 times the bandgap. Much lower thresholds are predicted and confirmed for QDs because of the relaxation of the requirement to conserve crystal momentum. Very efficient MEG has been experimentally observed in PbSe, PbS, PbTe, CdTe, and InAs QDs; threshold photon energies as low as $2E_g$ have been reported. Up to 3 electrons per photon (300% QY) have been observed at sufficiently high photon energies ($>4E_g$). A new model based on a coherent superposition of multiexcitonic states has been introduced to explain these results, but other models for MEG have also been proposed. Two excitons/photon yield over 90% of the efficiency benefit of MEG. Various configurations of QD solar cells are possible that could yield high conversion efficiencies; these include: nanocrystalline TiO₂ sensitised with QDs, QD arrays that exhibit 3-D miniband formation used as the *i*-region in *p-i-n* structures, and QDs embedded in a blend of electron- and hole-conducting phases (for example, conducting polymers). The dynamics of hot exciton cooling, forward and inverse Auger recombination (MEG), and electron transfer can be modified and controlled in QD solar cells to potentially allow very efficient solar photon conversion via efficient MEG. To date the efficiency of QD solar cells is rather low (<2%), but with further research and development it is expected the efficiency of such cells will improve markedly and lead to low-cost solar power

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